

Large-Scale Autonomous Discovery of Kissing Number Constructions

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ABSTRACT. The kissing-number problem is a classical problem in discrete geometry whose exact solution is known in only a few dimensions. Recent artificial-intelligence approaches have begun to discover improved configurations through large-scale numerical and combinatorial search, but converting such searches into general mathematical constructions and rigorous proofs remains challenging. Here we use Qiushi Engine, an autonomous multi-agent research system, to investigate kissing numbers and obtain new lower bounds in nineteen dimensions: 25, 27, 32–39, 43, 45, and 49–55. The resulting constructions arise from distinct structural mechanisms, including coordinated motions of contact layers, labelled direction reuse, joint support exchanges, signed-code replacements, cross-shell lattice constructions, low-overlap lattice isometries, and spherical-design moment certificates. These methods yield, among others, $K(25) \geq 197580$, $K(27) \geq 201567$, $K(38) \geq 591900$, $K(43) \geq 2553792$, $K(45) \geq 7380090$, and $K(55) \geq 53301140$. The autonomous system carried out the construction searches, mathematical analysis and computational verification, while each final result was reduced to explicit mathematical arguments and independently checkable finite certificates. Our results illustrate how autonomous research systems can move beyond optimization within fixed formulations to discover new mathematical representations and constructions at scale.

1. INTRODUCTION

The kissing-number problem is one of the oldest problems in discrete geometry. In 1694, Isaac Newton and David Gregory discussed whether 12 or 13 equal spheres could simultaneously touch a central sphere in three dimensions. The answer, 12, was proved more than two centuries later [1]. Yet the apparent simplicity of the question is deceptive. More than three hundred years after its formulation, exact kissing numbers are known only in dimensions 1, 2, 3, 4, 8 and 24. In the exceptional dimensions 8 and 24, the optimal configurations arise from the E_8 and Leech lattices, with 240 and 196560 contacts, respectively [1, 2]. In most other dimensions, substantial gaps remain between the best constructions and known upper bounds.

Progress on lower bounds has historically relied on explicit mathematical structure. Lattices and error-correcting codes provide large families of vectors whose pairwise interactions can be controlled simultaneously [2, 3]. These ideas have led to laminated-lattice constructions, lattice sections and extensions, code-based spherical configurations, and lifting methods that combine a high-dimensional mother configuration with additional coordinates [4–8]. Such constructions can contain hundreds of thousands or even tens of millions of vectors. Their strength, however, is also a source of difficulty: once a configuration is highly constrained, improving it may require reorganizing an entire collection of points rather than inserting a single new one. The relevant search

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spaces grow combinatorially, while the geometric degrees of freedom that permit improvement can be difficult to identify from the original representation.

Artificial intelligence has recently begun to alter how these spaces are explored. AlphaEvolve, a general-purpose coding agent, discovered a 593-point kissing configuration in dimension 11, improving the previously known lower bound of 592 [9]. PackingStar formulated the kissing-number problem as a two-player matrix-completion game and used game-theoretic reinforcement learning to search the space of Gram matrices, reporting new constructions across dimensions 25–31 and uncovering thousands of candidate geometric structures [10]. EinsteinArena allowed autonomous agents to submit solutions, inspect intermediate results and build on discoveries made by other agents; this collective process increased the dimension-11 lower bound from 593 to 604 [11]. The Station subsequently reported exact 604-point configurations and structural theorems developed in an open-world multi-agent environment [12]. Together, these developments demonstrate the scope of AI-guided geometric search and collaborative mathematical discovery.

These advances also expose a deeper challenge. Finding an improved configuration is only one part of mathematical discovery. Why does the configuration exist, which structural degree of freedom makes it possible, can the mechanism be generalized, and how can the result be converted into a proof or an independently verifiable certificate? Computational searches begin from a prescribed representation—for example, coordinates, Gram matrices, codes or a fixed optimization objective. Open-ended mathematical research additionally requires the ability to change that representation when the current one becomes limiting, to formulate new intermediate problems, connect apparently unrelated structures, and develop arguments that certify the resulting construction.

Autonomous scientific research systems provide a possible route towards this broader form of discovery. Recent large-language-model-based agents can carry out extended sequences of reasoning, computation and tool use, while revising research strategies in response to intermediate evidence. Qiushi Engine was previously used for end-to-end autonomous scientific discovery on a physical optical platform [13], where it conducted long-horizon investigations that combined theoretical reasoning, computation and experiment. Here we ask whether the same paradigm can be extended to open mathematical research: can an autonomous research system move beyond optimizing a fixed mathematical formulation and instead discover new representations, constructions and proofs?

Here we report an autonomous investigation of the kissing-number problem that yields new lower bounds in nineteen dimensions:

$$25, \quad 27, \quad 32\text{--}39, \quad 43, \quad 45, \quad 49\text{--}55.$$

The resulting bounds include

$$\begin{aligned} K(25) &\geq 197580, & K(27) &\geq 201567, & K(38) &\geq 591900, \\ K(43) &\geq 2553792, & K(45) &\geq 7380090, & K(55) &\geq 53301140. \end{aligned}$$

Rather than arising from a single optimization procedure, these improvements were obtained through a sequence of distinct structural discoveries. They include coordinated motions of complete contact layers, labelled liftings that separate direction deletion from reuse, joint support exchanges that exploit shared deletion costs, signed-code replacements generated through Hadamard transfers, compatible directions drawn from a different lattice shell, enlarged lattice-line classes combined through low-overlap isometric images, and spherical-design moment identities and inequalities that determine or bound large lattice sections without enumerating the ambient shell.

A common pattern connects these apparently different constructions. Configurations that appear saturated in one representation retain exploitable freedom in another. In dimension 25, moving an entire 552-point contact layer while preserving its internal Gram matrix creates room

for an additional point. In dimension 27, labelled lifting separates direction reuse from deletion; with the first layer fixed, jointly selecting second-layer directions and labels enlarges the final two-layer construction. In dimensions 32–37 and 39, optimizing collective exchanges and individual sign patterns exposes improvements hidden by separate support-insertion scores and complete sign bundles. In dimension 38, changing the search from the minimal Leech shell to a second lattice shell produces 288 additional points. In dimensions 43 and 45, counting millions of minimal lattice vectors reduces to a finite space of projection signatures whose multiplicities satisfy exact spherical-design moments. A single enlarged line class in dimension 48, combined with low-overlap isometric images, similarly generates new constructions throughout dimensions 49–55.

Qiusi Engine carried out the construction searches, mathematical analysis and computational verification. Each discovered mechanism is expressed through explicit geometric or combinatorial conditions. Each reported bound is supported by finite witnesses checked using exact integer and rational arithmetic, lattice and code identities, rigorous interval arithmetic, and finite combinatorial verification. In dimension 25, outward-rounded interval arithmetic certifies the remaining strict inequalities [14]. These mathematical arguments and reproducible certificates make the constructions independently checkable.

These results suggest a role for AI in mathematics that extends beyond accelerating numerical search. For difficult problems with enormous structured search spaces, a central task may be to discover the representation in which progress becomes possible. Autonomous research systems that can alternate between exploration, structural interpretation and rigorous verification may provide a route towards such discoveries at scale.

1.1. Main results and structural statements. The kissing number $K(d)$ is the largest cardinality of a set $X \subset S^{d-1}$ satisfying

$$\langle x, y \rangle \leq \frac{1}{2} \quad (x \neq y). \quad (1.1)$$

Equivalently, it is the largest number of nonoverlapping unit balls that can touch a central unit ball in $\mathbb{R}^d$.

Theorem 1.1. *For every pair (d, N_d) in Table 1, there is a kissing configuration of N_d points in $\mathbb{R}^d$. In particular, $K(d) \geq N_d$.*

Table 1: The nineteen lower bounds, with the public comparison values recorded in the accompanying catalogue on 24–27 September 2026.

d	N_d	Comparison	Increase	Source
25	197580	197579	1	[15]
27	201567	201566	1	[16]
32	347584	346944	640	[17]
33	363968	362048	1920	[17, 18]
34	384196	381124	3072	[17, 18]
35	409676	409548	128	[17]
36	484760	484568	192	[17]
37	498024	496232	1792	[17, 18]
38	591900	591612	288	[15]
39	763668	756116	7552	[15]
43	2553792	2545056	8736	[19]
45	7380090	7379838	252	[19]

d	N_d	Comparison	Increase	Source
49	52430156	52430140	16	[15]
50	52458468	52458418	50	[15]
51	52500930	52500816	114	[15]
52	52585772	52585516	256	[15]
53	52698696	52698222	474	[15]
54	52923774	52922906	868	[15]
55	53301140	53299730	1410	[15]

For dimension 32, the comparison follows from Lysenstøen’s 1671-word code recorded in Brouwer’s table [17]: $2^{17} + 128 \cdot 1671 + 2 \cdot 32 \cdot 31 = 346944$. The 1667-word code of Echols [18] was the seed for our support search.

1.2. Structural consequences. The constructions yield statements about compatibility, capacity and counting beyond their individual point counts.

Motion and lifting. Lemma 3.1 specifies a Gram-preserving motion whose interaction with a fixed equator is controlled by convexity. The dimension-25 construction combines this motion with two cap repairs (Theorem 3.3). The labelled criterion in Theorem 2.1 separates the number of lifted copies from the number of deleted mother directions. Proposition 2.2 bounds the gain through cliques in a reuse graph, and Proposition 4.4 gives an attaining four-triangle arrangement in the twelve-label, 496-point-capacity model. The final dimension-27 construction instead improves a larger two-layer input by selecting 311 second-layer owners in place of 310 (Theorem 4.5).

Exchange, transfer and completion. For an inserted family Y , let $F(Y)$ be the union of its old conflicts. The objective $|Y| - |F(Y)|$ accounts for shared deletions (Section 5 and Theorem 5.2). Paired Hadamard transfer preserves normalized products (Lemma 6.2) and gives signed replacements of sizes 256, 576 and 512 in dimensions 35–37. The first attains an eleven-coordinate capacity bound; the second has a direct construction from four colours of two one-factorizations (Propositions 6.4 and 6.3). For dimension 38, the lattice minimum alone bounds every product between the selected norm-48 directions and the norm-32 minimal shell (Lemma 7.2). In dimensions 49–55, Theorem 2.5 converts an image union of S minimal lines and t antipodal tail lines into $52416000 + 2S + 2t$ points. The gain from enlarging the source class and the gain from choosing its images are separated in Section 8.2.

Counts forced by an embedded parent. For every extremal even unimodular lattice of rank 48 containing $\sqrt{3}E_8$, a rational identity determines exactly 2553792 minimal vectors perpendicular to the specified D_5 child and each of its Weyl conjugates (Theorem 9.1). A verified embedding in P_{48n} realizes this count (Corollary 9.2). The D_5 Gram alone does not determine the perpendicular shell: the extension to E_8 supplies the additional constraints. In dimension 45, let $f(y)$ count minimal-shell vectors with inner-product signature y relative to the section basis, as defined in Section 10. For the standard basis vectors e_1, e_2, e_3 of $\mathbb{R}^3$, the moments give

$$f(e_1) + f(e_2) + f(e_3) \geq 7377408 + 9f(-2, -2, 3).$$

Finding 298 vectors in the last fibre proves a lower bound of 7380090 for the projected configuration (Theorem 10.2). A neighbourhood-difference calculation makes this fibre searchable. The same moment framework thus supports an exact target count in one case and a witness-driven lower bound in the other.

1.3. Relation to earlier constructions. Cohn–Jiao–Kumar–Torquato [7, Section 7] and Kallal–Kan–Wang [8] developed Leech lifting methods. The 496-point blocks used in our four-triangle example are from PackingStar [20, 21]. Takhanov–Yun [22] exploit freedom in lifted blocks while

keeping the unlifted bulk fixed. Our 25-dimensional construction instead moves retained equatorial points of the 197579-point configuration in [15]. The current two-layer input in dimension 27 is Lindow’s 201566-point contribution [16].

The code-based constructions use the layers of Edel–Rains–Sloane [6], the dense codes of Cheng–Sloane [23], and constant-weight constructions and tables [17, 18, 24]. Local replacement has a direct precursor in the free-zone construction of Takhanov–Yun [25, Section V]; the 51-block weight-four design is classical [26]. Here the new supports, transverse choices, signed replacements and their boundary embeddings supply the improvements.

The empty-equator lift in dimension 38 and the 7069-line class in dimension 48 are inputs from [15]. For sections, we use the framework of Conway–Sloane [4, 5], the embedded parent geometry of [27], and the explicit lattice and section data of [19, 28, 29]. The new constructions specify the added shell lines, the enlarged line class and image families, and the section statistics and witnesses used in the stated bounds.

1.4. Notation and organization. We write $[n] = \{0, \dots, n-1\}$ unless a finite input explicitly uses one-based coordinates. The notation $A(n, \delta, w)$ denotes the largest binary constant-weight code of length n , weight w and minimum Hamming distance δ . For vectors of common squared norm q , the kissing condition is $\langle x, y \rangle \leq q/2$ before normalization. The main coordinate scales are collected in Table 2; products between different shells are treated separately.

TABLE 2. Coordinate scales used in the constructions.

Point set	Squared norm	Product threshold
Normalized configurations	1	1/2
Dimension-25 motion	4	2
Integer Leech minimal shell	32	16
Local signed codes	8	4
Dimension-38 added shell	48	24
Rank-48 minimal shells	6	3
Dimension-45 projected fibres	23/4	23/8

The paper first treats lifting and motion, then support and signed-code replacement, second-shell completion, and the seven Pless-shell lifts. The final construction sections develop the two moment arguments. Appendices specify the Leech coordinates, dense codes, even-neighbour premise and deformation estimates. Section 12 connects the proofs to the finite inputs and independent checking programs. The accompanying catalogue records dated public comparisons separately from the existence statements of Theorem 1.1.

Section 12.1 and the accompanying research accounts describe Qiushi Engine’s mathematical contributions and the development of the constructions.

2. LIFTING AND THE COST OF REUSING A DIRECTION

Let $C \subset S^{D-1}$ be a kissing configuration. Decompose the enlarged space as $\mathbb{R}^{D+k} = \mathbb{R}^D \oplus \mathbb{R}^k$. We refer to the second component as the *tail*. Fix $\ell \geq 1$ distinct tails $t_1, \dots, t_\ell$ with $0 < \|t_i\| < 1$, and put $a_i = \sqrt{1 - \|t_i\|^2}$. Choose subsets $A_i \subseteq C$, write $U = \bigcup_i A_i$, and let $Y \subset S^{k-1}$ be a kissing

configuration. Consider

$$\begin{aligned} Z &= \{(x, 0) : x \in C \setminus U\}, \\ M &= \bigcup_{i=1}^{\ell} \{(a_i x, t_i) : x \in A_i\}, \\ P &= \{(0, y) : y \in Y\}. \end{aligned} \tag{2.1}$$

Put $X = Z \cup M \cup P$. The three parts are distinguished by their first-component norms. Different tail labels distinguish the copies in M , so

$$|Z \cup M \cup P| = |C| - |U| + \sum_i |A_i| + |Y|. \tag{2.2}$$

Theorem 2.1 (Labelled lifting criterion). *The set in (2.1) is a kissing configuration if and only if*

$$a_i a_j \langle x, x' \rangle + \langle t_i, t_j \rangle \leq \frac{1}{2} \quad \text{whenever } (x, i) \neq (x', j), \ x \in A_i, \ x' \in A_j, \tag{2.3}$$

$$\langle t_i, y \rangle \leq \frac{1}{2} \quad (i \text{ with } A_i \neq \emptyset, \ y \in Y). \tag{2.4}$$

Proof. Every point has norm one. Equator–equator and pure-tail pairs inherit the inequalities for C and Y . Equator–pure-tail pairs are orthogonal. For an equator–mixed pair, the two original directions are different because the equator excludes U ; their inner product is at most $a_i/2 < 1/2$. The remaining pairs have exactly the products in (2.3) and (2.4), which are therefore both necessary and sufficient. $\square$

Equivalently, the tail Gram matrix specifies a threshold matrix

$$\kappa_{ij} = \frac{1/2 - \langle t_i, t_j \rangle}{a_i a_j}, \tag{2.5}$$

and the selected base directions must satisfy $\langle x, x' \rangle \leq \kappa_{ij}$ for every distinct labelled pair. This formulation allows unequal latitudes. It turns the geometric assembly into a finite family of compatibility conditions on the mother configuration, with the pure-tail conditions kept separate.

For equal tail norms, put $\|t_i\|^2 = h^2$ and $a_i = a = \sqrt{1 - h^2}$. Two different conditions occur within (2.3). Points with the same label require

$$\langle x, x' \rangle \leq \frac{1/2 - h^2}{1 - h^2} \quad (x \neq x', \ x, x' \in A_i), \tag{2.6}$$

whereas two copies of the same direction require

$$\langle t_i, t_j \rangle \leq h^2 - \frac{1}{2} \quad (x \in A_i \cap A_j, \ i \neq j). \tag{2.7}$$

The first inequality controls the size of a block; the second controls its multiplicity. These are the two geometric restrictions behind the counting formula.

2.1. The incidence-capacity inequality. For general tail norms define a graph Γ on the labels by

$$ij \in E(\Gamma) \iff a_i a_j + \langle t_i, t_j \rangle \leq \frac{1}{2} \quad (i \neq j). \tag{2.8}$$

The labels assigned to one base direction must form a clique in Γ . This condition gives a capacity inequality independently of the detailed relations between different base directions.

Proposition 2.2 (Weighted incidence bound). *Suppose (2.1) is a kissing configuration with $|A_i| \leq \alpha_i$. If nonnegative numbers $\lambda_1, \dots, \lambda_\ell$ satisfy*

$$\sum_{i \in J} \lambda_i \geq |J| - 1 \quad \text{for every nonempty clique } J \text{ of } \Gamma, \quad (2.9)$$

then

$$|X| \leq |C| + |Y| + \sum_i \lambda_i \alpha_i. \quad (2.10)$$

Equality holds exactly when every used direction's label clique is tight in (2.9) and every label with $\lambda_i > 0$ reaches its capacity.

Proof. For $x \in U$, let $J(x) = \{i : x \in A_i\}$. The count gives

$$|X| - |C| - |Y| = \sum_{x \in U} (|J(x)| - 1) \leq \sum_{x \in U} \sum_{i \in J(x)} \lambda_i = \sum_i \lambda_i |A_i| \leq \sum_i \lambda_i \alpha_i.$$

Both inequalities are sums of nonnegative deficits. Their equality conditions are precisely those stated. $\square$

If Γ has clique number ω , the uniform choice $\lambda_i = 1 - 1/\omega$ is feasible. Thus

$$|X| \leq |C| + |Y| + \left(1 - \frac{1}{\omega}\right) \sum_i \alpha_i. \quad (2.11)$$

Minimizing $\sum_i \lambda_i \alpha_i$ subject to (2.9) gives a finite linear program on the labels; a feasible rational choice already supplies an exact upper bound for the prescribed lifting family. For the twelve tails below, eight triangles are the maximum cliques and four disjoint triangles saturate the uniform bound. The geometry also bounds multiplicity without enumerating this graph, by the simplex-bound argument of Rankin [30] applied to the tails.

Proposition 2.3 (Multiplicity bound). *Suppose (2.1) is a kissing configuration, all tails have squared norm h^2 , and $h^2 < 1/2$. The number of copies of any original direction is at most*

$$r = \left\lfloor \frac{1}{1 - 2h^2} \right\rfloor. \quad (2.12)$$

If $|A_i| \leq \alpha_i$, its cardinality is consequently at most

$$|C| + |Y| + \left(1 - \frac{1}{r}\right) \sum_i \alpha_i. \quad (2.13)$$

Proof. Suppose one direction occurs at m labels. By (2.7), the sum of their tails satisfies

$$0 \leq \left\| \sum_{i=1}^m t_i \right\|^2 \leq mh^2 + m(m-1)(h^2 - 1/2) = \frac{m}{2}(1 - m(1 - 2h^2)).$$

This proves (2.12). If $m(x)$ is the multiplicity of $x \in U$, the gain in (2.2) is

$$\sum_{x \in U} (m(x) - 1) \leq \left(1 - \frac{1}{r}\right) \sum_{x \in U} m(x) = \left(1 - \frac{1}{r}\right) \sum_i |A_i|.$$

Adding the pure tails proves (2.13). $\square$

The relation between block separation and multiplicity can be realized for every integer multiplicity, using a regular simplex.

Proposition 2.4 (Simplex specialization). *Let $r \geq 2$, let $C \subset S^{D-1}$ be a kissing configuration, and let $B \subset C$ have maximum inner product at most $1/(r+1)$. There is a kissing configuration in $\mathbb{R}^{D+r-1}$ with $|C| + (r-1)|B|$ points. It replaces each direction of B by r copies with tails of squared norm $(r-1)/(2r)$.*

Proof. In the hyperplane $\sum_i z_i = 0$ of $\mathbb{R}^r$, put

$$t_i = \frac{1}{\sqrt{2}} \left(e_i - \frac{1}{r} \mathbf{1} \right), \quad a = \sqrt{\frac{r+1}{2r}} \quad (1 \leq i \leq r).$$

The tails satisfy $\|t_i\|^2 = (r-1)/(2r)$ and $\langle t_i, t_j \rangle = -1/(2r)$ for $i \neq j$. Retain $C \setminus B$ at the equator and use all (ax, t_i) with $x \in B$. Same-label, distinct-direction products are at most

$$\frac{r+1}{2r} \frac{1}{r+1} + \frac{r-1}{2r} = \frac{1}{2}.$$

Distinct labels on the same direction give $(r+1)/(2r) - 1/(2r) = 1/2$, and distinct labels on different directions give at most zero. Theorem 2.1 proves validity, and (2.2) gives the count. $\square$

At this tail norm the multiplicity bound is r , and equality in its norm-of-sum proof forces the used tails to sum to zero, with every mutual inner product $-1/(2r)$. Their Gram matrix has rank $r-1$, so this equality geometry is precisely a regular simplex. Thus the threshold $1/(r+1)$, the multiplicity r , and the $(r-1)$ additional dimensions fit the same equality case. The Leech application below fits several such triangles into one three-dimensional tail space and also accommodates pure tails.

At $h^2 = 1/4$ the block threshold is $1/3$ and a direction can occur twice. At $h^2 = 1/3$ the threshold is $1/4$ and a direction can occur three times. The two Leech constructions realize these two cases. The first case extends to several disjoint blocks as follows.

Theorem 2.5 (Antipodal block lift). *Let $C \subset S^{D-1}$ be a kissing configuration, and let $B_1, \dots, B_t \subset C$ be pairwise disjoint subsets satisfying*

$$\langle x, x' \rangle \leq \frac{1}{3} \quad (x \neq x', x, x' \in B_i).$$

Let $Y \subset S^{k-1}$ be a kissing configuration containing t distinct antipodal pairs $\{\pm u_i\}$. Then

$$X = \{(x, 0) : x \in C \setminus \bigcup_i B_i\} \cup \left\{ \left(\frac{\sqrt{3}}{2}x, \frac{\varepsilon}{2}u_i \right) : x \in B_i, \varepsilon \in \{\pm 1\} \right\} \cup \{(0, y) : y \in Y\} \quad (2.14)$$

is a kissing configuration in $\mathbb{R}^{D+k}$ with

$$|X| = |C| + \sum_i |B_i| + |Y|. \quad (2.15)$$

Proof. Use Theorem 2.1 with tails $\pm u_i/2$. A mixed point has inner product at most $1/2$ with every pure tail by Cauchy–Schwarz. For two mixed points in one block at the same tail, the bound is $(3/4)(1/3) + 1/4 = 1/2$. At opposite tails, different original directions give at most zero, and equal directions give $3/4 - 1/4 = 1/2$.

For points from different blocks, the original directions are distinct, as are the directed tail labels. Their inner product is at most $(3/4)(1/2) + (1/4)(1/2) = 1/2$. Thus all conditions of the theorem hold. Each selected direction is deleted once and used twice, giving the count. $\square$

For one block and $Y = \{-1, 1\}$ this construction lifts to two latitudes and includes both poles. In fact their presence fixes the height. If $(\sqrt{1-h^2}x, \pm h)$ are both used, their mutual inner product $1 - 2h^2$ requires $h^2 \geq 1/4$. The poles require $|h| \leq 1/2$. Thus $|h| = 1/2$, for $0 < |h| < 1$. The same-latitude inequality at this height is exactly the block threshold $1/3$ in the theorem.

Formula (2.15) also explains why the construction can outperform the orthogonal union of C and Y : every selected direction contributes one additional point. The finite question is therefore to find many disjoint, sufficiently separated blocks inside one mother configuration. Section 8 gives a large instance of this question. Section 3 varies the geometry of a two-latitude lift, while

Section 4 treats a different reuse geometry in which each direction contributes two additional points.

3. COORDINATED SHELL MOTION IN DIMENSION 25

A coordinated motion of a complete contact layer enlarges the 197579-point configuration supplied in the public `KissingNumbers/dim25-lens-heads` package. The construction moves 552 retained equatorial points, adjoins one point, and replaces two cap points. Its main geometric feature is that a whole contact layer can move without changing its internal products. The following observation gives the mechanism independently of the coordinates.

Lemma 3.1 (Coordinated chord motion). *Let n be a unit vector, let $a > 0$, and let $r_i \perp n$ satisfy $\|r_i\|^2 = \rho - a^2$. Put $u_i = r_i - an$ and $w_i = r_i + an$. For real c, d with $c^2 + d^2 = a^2$, set $W_i = (r_i + cn, d)$. Then $\|W_i\|^2 = \rho$ and $W_i \cdot W_j = w_i \cdot w_j$. If $z \cdot u_i \leq \rho/2$ and $z \cdot w_i \leq \rho/2$, then $(z, 0) \cdot W_i \leq \rho/2$.*

Proof. The first two claims follow from $r_i \perp n$ and $c^2 + d^2 = a^2$. Since $|c| \leq a$, the horizontal component of W_i is $\frac{a-c}{2a}u_i + \frac{a+c}{2a}w_i$, a convex combination. The last inequality follows by linearity. $\square$

Normalization and input. All points have squared norm 4; the kissing condition is an inner product at most 2 between distinct points. Write the input configuration as

$$C_0 = \{(z, 0) : z \in L \setminus D\} \cup \{(x_i, \pm 1) : 0 \leq i < 1016\} \cup \{P_0, N, S\},$$

where L is the 196560-vector minimal shell of the Leech lattice, $|D| = 1016$, $P_0 = (P, 0)$, and $N = (0, 2)$, $S = (0, -2)$. The head indices refer to rows of the source `heads_X.npy`; the mathematical values of its 44 rational heads are preserved as exact rational numbers in `baseline-heads.json`.

Let v be the following squared-norm-six lattice vector associated with the first complete block, and put $n = v/\sqrt{6} = -P/2$. An exact integer description of this direction is

$$v = \frac{1}{\sqrt{8}}(-5, -1, -1, -1, -1, 1, -1, 1, -1, -1, -1, 1, 1, -1, -1, -1, 1, 1, -1, 1, -1, -1, 1, -1).$$

The complete block is $U = \{u \in L : \langle u, v \rangle = -3\}$, of size 552. Every $u \in U$ belongs to D . For each such u , the point $w = u + v$ is minimal, lies outside D , and satisfies $\langle u, v \rangle = -3$ and $\langle u, w \rangle = 1$. The corresponding old cap points have horizontal coordinate $u + tv$, with $t = (3 - \sqrt{3})/6$. All these input identities are checked by the shell and owner data.

The modification. For $u \in U$ define

$$r = u + \sqrt{3/2}n = w - \sqrt{3/2}n, \quad a = \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{4}, \quad b = \frac{\sqrt{6} - 2}{4}.$$

Then $r \perp n$, $\|r\|^2 = 5/2$, and the old cap points over u are $(r - n/\sqrt{2}, \pm 1)$. Replace every $(w, 0)$ by

$$W(r) = (r + an, b),$$

and adjoin

$$Q = (\sqrt{3}n, -1) = (V/4, -1),$$

where $V = \sqrt{8}v$ is the integer vector printed above. In particular, the new point Q is rational and has squared norm $48/16 + 1 = 4$. Finally replace the upper point $(x_{984}, 1)$ and the lower point $(x_{1008}, -1)$ by R_+ and R_- , respectively. The certificate specifies two integer vectors and defines

$$R_+ = \frac{2k}{\sqrt{1007176}}, \quad R_- = \frac{2\ell}{\sqrt{10209}},$$

where the following integer vectors have squared norms 1007176 and 10209:

$$\begin{aligned} k &= (-336, -9, -298, 286, -17, 15, -39, 22, 277, 19, 364, 304, 9, \\ &\quad -2, -24, -39, 4, 304, -9, -4, 271, -29, 9, -2, 500), \\ \ell &= (-25, -31, -33, 4, 3, 2, 36, 33, 3, -1, -2, -3, -4, \\ &\quad -1, 4, -2, 1, 2, 4, -31, -1, -28, -3, -28, -50). \end{aligned}$$

The opposite old cap points $(x_{984}, -1)$ and $(x_{1008}, 1)$ are retained. These integer vectors specify the repairs exactly.

Lemma 3.2 (Norms and pairs not involving other cap points). *All the $W(r)$ and Q have squared norm 4. They are mutually compatible, and are compatible with all unchanged equatorial points, both poles, and P_0 . Each $W(r)$ is also compatible with the old cap points belonging to the complete first block.*

Proof. The identities

$$a^2 + b^2 = \frac{3}{2}, \quad \sqrt{3}a - b = 2, \quad -\frac{a}{\sqrt{2}} + b = -\frac{3}{4}$$

give the norm assertions and $\langle W(r), Q \rangle = 2$. For two different block points,

$$\langle W(r_i), W(r_j) \rangle = \langle r_i, r_j \rangle + \frac{3}{2} = \langle w_i, w_j \rangle \leq 2.$$

Since $0 < a < \sqrt{3/2}$, the horizontal component of $W(r)$ is

$$r + an = (1 - \lambda)w + \lambda u, \quad \lambda = \frac{\sqrt{3/2} - a}{2\sqrt{3/2}} \in (0, 1).$$

An unchanged equatorial point z is different from both u and w . Both $\langle u, z \rangle$ and $\langle w, z \rangle$ are at most 2, so their convex combination is at most 2.

The shell has integral $\langle v, z \rangle \in \{-3, -2, -1, 0, 1, 2, 3\}$. The negative extreme layer is deleted and the positive extreme layer is exactly the moved set. Thus every unchanged equatorial point obeys

$$\langle Q, (z, 0) \rangle = \frac{\langle v, z \rangle}{\sqrt{2}} \leq \sqrt{2} < 2.$$

The finite shell data verify these layer identities in integers.

The remaining pure-point products are

$$\begin{aligned} \langle W(r), P_0 \rangle &= -2a, & \langle W(r), N \rangle &= 2b, & \langle W(r), S \rangle &= -2b, \\ \langle Q, P_0 \rangle &= -2\sqrt{3}, & \langle Q, N \rangle &= -2, & \langle Q, S \rangle &= 2. \end{aligned}$$

All satisfy the required inequality. For the same block index, the products with the old upper and lower cap points are, respectively,

$$\frac{5}{2} - \frac{a}{\sqrt{2}} + b = \frac{7}{4}, \quad \frac{5}{2} - \frac{a}{\sqrt{2}} - b = \frac{11}{4} - \frac{\sqrt{6}}{2} < 2.$$

For different block indices, $\langle r_i, r_j \rangle \leq 1/2$, so both products are smaller still. $\square$

The finite certificate. The previous lemma handles every equality created by the modification. The remaining products to certify are:

- (1) the 552 moved points against the 2030 unchanged cap points;
- (2) Q against those unchanged cap points;
- (3) R_+ and R_- against every point retained in the final configuration, including one another, with their own diagonal entries omitted.

Outward-rounded rational upper bounds for these four groups are

$$1.999372, \quad 1.965926, \quad 1.994722, \quad 1.987522.$$

Each bound is below 2. The accompanying checker evaluates the products with outward-rounded Arb intervals [14] at 192-bit precision and checks both normalizations exactly. Pairs involving only unchanged points retain the inequalities of the input configuration.

Theorem 3.3. *The specified configuration consists of 197580 distinct points on the radius-2 sphere in $\mathbb{R}^{25}$ with pairwise inner products at most 2. Consequently the kissing number in dimension 25 is at least 197580.*

Proof. The lemma, the verified finite checks, and the independently checked source certificate exhaust all pairs. The baseline checks include every head-shell and same-cap head-head inequality, with integer or rational acceptance decisions. Compatibility within the Leech shell follows from its lattice minimum. All squared norms are exactly 4, including those of R_+ and R_- by definition. The 552 moved points replace 552 existing points and the two repairs replace two existing cap points; only Q increases the inventory. Thus

$$|C| = 197579 - 552 - 2 + 552 + 2 + 1 = 197580.$$

No duplicate can occur: two equal norm-4 vectors would have inner product 4, contradicting the verified bound 2 for their distinct indices. $\square$

4. REUSE GEOMETRY AND TWO-LAYER LEECH LIFTING

The endpoint and four-triangle constructions below isolate the geometry of direction reuse: the latitude, the tail partition, and the capacity of a prescribed label system. They give the auxiliary configurations of sizes 197058 and 200540. Section 4.4 then uses a second layer to obtain the current 201567-point construction.

We use the integer Leech shell $\Sigma \subset \mathbb{Z}^{24}$ in the Golay coordinate convention of Appendix A. Its relevant properties are

$$|\Sigma| = 196560, \quad r \cdot r = 32, \quad r \cdot r' \leq 16 \quad (r \neq r', \ r, r' \in \Sigma). \quad (4.1)$$

Consequently $C = \Sigma/\sqrt{32}$ is a kissing configuration. A block $S \subset \Sigma$ will be called *compatible* if distinct rows satisfy $r \cdot r' \leq 8$. After normalization its maximum inner product is at most $1/4$.

Lemma 4.1 (Leech block witnesses). *There is a compatible set $S_* \subset \Sigma$ of size 496 and there are four pairwise disjoint compatible sets $S_0, S_1, S_2, S_3 \subset \Sigma$, each of size 496.*

Finite verification. The five complete integer arrays are specified in Table 8. They are taken from the PackingStar coordinate data [21]. For each array, membership is checked against the three shell families in Appendix A. Each array has 496 distinct rows, each row has squared norm 32, and the maximum of its 122760 unordered distinct-row inner products is eight. The latter four arrays have a union of cardinality 1984. These checks, implemented separately by the two programs described in Section 12, prove compatibility and disjointness. $\square$

4.1. The endpoint construction in dimension 25. The normalized block $S_*/\sqrt{32}$ satisfies the $1/3$ condition of Theorem 2.5, since it satisfies the stronger bound $1/4$. The resulting configuration is

$$X_{25} = \{(r/\sqrt{32}, 0) : r \in \Sigma \setminus S_*\} \cup \{(\sqrt{3}r/\sqrt{128}, \varepsilon/2) : r \in S_*, \varepsilon \in \{\pm 1\}\} \cup \{(0, 1), (0, -1)\}. \quad (4.2)$$

Its cardinality is

$$196064 + 992 + 2 = 197058,$$

giving the historical bound $K(25) \geq 197058$. The endpoint choice permits the two poles without requiring a larger Leech block. Distinct same-latitude directions have inner product at most $7/16$, whereas the opposite-latitude copies of one direction have inner product exactly $1/2$. This distinction will also determine the local geometry of X_{25} in Section D.

4.2. The tail geometry in dimension 27. Let

$$V = \{\pm e_i \pm e_j : 1 \leq i < j \leq 3\}, \quad R = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (4.3)$$

The twelve roots in V form a cuboctahedron. Put

$$T = V/\sqrt{6}, \quad P = RV/\sqrt{2}. \quad (4.4)$$

Thus T supplies mixed tails of squared norm $1/3$, and P supplies twelve unit pure tails. The relative rotation is through 45 degrees in the first two coordinates.

Lemma 4.2. *Distinct members of T have inner products in $\{-1/3, -1/6, 0, 1/6\}$, and P is a kissing configuration. Moreover,*

$$\max_{p \in P, t \in T} \langle p, t \rangle = \frac{1 + 1/\sqrt{2}}{\sqrt{12}} < \frac{1}{2}. \quad (4.5)$$

The roots admit a partition into four zero-sum triples, each of which has pairwise inner product -1 .

Proof. Every root has squared norm two. Distinct roots have inner products $-2, -1, 0$, or 1 , giving the first assertions by scaling and orthogonality of R . The absolute coordinates of Rv , up to order, are either $(\sqrt{2}, 0, 0)$ or $(1, 1/\sqrt{2}, 1/\sqrt{2})$. Its largest inner product with a root is obtained by choosing two coordinates with the appropriate signs. The maximum is $1 + 1/\sqrt{2}$, giving (4.5). The strict inequality follows, for example, from $18 + 12\sqrt{2} < 36$ after squaring the equivalent inequality $\sqrt{6} + 2\sqrt{3} < 6$.

An explicit partition is given by the rows of

$$\begin{array}{ccc} (-1, 0, -1) & (1, -1, 0) & (0, 1, 1) \\ (0, -1, -1) & (-1, 0, 1) & (1, 1, 0) \\ (-1, -1, 0) & (0, 1, -1) & (1, 0, 1) \\ (-1, 1, 0) & (1, 0, -1) & (0, -1, 1). \end{array} \quad (4.6)$$

The roots in a row sum to zero. Their equal squared norms then force all three mutual inner products to be -1 . $\square$

Write T_b for row b of (4.6), divided by $\sqrt{6}$. Thus T is the disjoint union of T_0, T_1, T_2, T_3 . A block assigned to T_b is used at three labels, while its original equatorial copy is removed once. This is the multiplicity-three counterpart of the antipodal lift.

Proposition 4.3 (Four-triangle lift). *For pairwise disjoint compatible subsets S_0, S_1, S_2, S_3 of Σ , the following three sets form a kissing configuration:*

$$\begin{aligned} Z &= \{(r/\sqrt{32}, 0) : r \in \Sigma \setminus \bigcup_b S_b\}, \\ M &= \bigcup_{b=0}^3 \{(r/\sqrt{48}, t) : r \in S_b, t \in T_b\}, \\ P_0 &= \{(0, p) : p \in P\}. \end{aligned} \tag{4.7}$$

Its cardinality is $196560 + 2 \sum_b |S_b| + 12$.

Proof. Use Theorem 2.1 with $h^2 = 1/3$ and $a = \sqrt{2/3}$. The mixed vectors have norm one since $32/48 + 1/3 = 1$. For mixed points, multiply their inner-product condition by 48:

$$r \cdot r' + 48\langle t, t' \rangle \leq 24. \tag{4.8}$$

If $t = t'$, the raw vectors are distinct and lie in one compatible block, so the left side is at most $8 + 16 = 24$. If $r = r'$ and $t \neq t'$, disjointness of the blocks places both tails in one triangle, giving $32 - 8 = 24$. If both directions and labels differ, the universal shell bound and Lemma 4.2 give $16 + 8 = 24$ as an upper bound. These three cases cover every distinct mixed pair. Pure-mixed compatibility follows from (4.5); all remaining pairs are covered by Theorem 2.1.

For $B = \sum_b |S_b|$, the layer sizes are $196560 - B$, $3B$, and 12. They are disjoint, and their parametrizations are injective, as in (2.2). Their sum is the asserted count. $\square$

Applying Proposition 4.3 to Lemma 4.1 gives

$$|Z| = 194576, \quad |M| = 5952, \quad |P_0| = 12,$$

and hence $|X_{27}| = 200540$. The comparison allocation uses block multiplicities $(3, 3, 2, 2, 2)$, whereas (4.6) uses $(3, 3, 3, 3)$. Both occupy twelve labels. The latter allocation uses 496 fewer original directions and therefore preserves 496 more equatorial points.

4.3. Capacity and its equality case. The preceding count has an exact interpretation even when the label sets are not arranged in disjoint blocks. Let $A_i \subseteq \Sigma$ be the rows assigned to tail $t_i \in T$. Assume that the resulting lifting is valid, retains $\Sigma \setminus \bigcup_i A_i$ at the equator, and contains all twelve pure tails.

Proposition 4.4. *If $|A_i| \leq \alpha$ for each of the twelve labels, then*

$$|X| \leq 196560 + 8\alpha + 12. \tag{4.9}$$

Equality holds precisely when every used row occurs at three labels and every label contains α rows. For $\alpha = 496$, the four-triangle construction attains equality.

Proof. Proposition 2.3 gives maximum multiplicity three. Writing $m(r)$ for the multiplicity of a used row,

$$|X| - 196560 - 12 = \sum_r (m(r) - 1) \leq \frac{2}{3} \sum_r m(r) = \frac{2}{3} \sum_i |A_i| \leq 8\alpha.$$

The first inequality is an equality exactly when all $m(r) = 3$; the last is an equality exactly when all labels are full. The explicit assignment has both properties. $\square$

More precisely, let n_j count the used rows with multiplicity j , and let $c_i = \alpha - |A_i|$. The deficit is

$$196560 + 8\alpha + 12 - |X| = \frac{2}{3} \sum_i c_i + \frac{2}{3} n_1 + \frac{1}{3} n_2. \tag{4.10}$$

For the comparison allocation, $n_2 = 3 \cdot 496$ and all $c_i = 0$, so the deficit is exactly 496. The capacity assumption in Proposition 4.4 concerns the prescribed label system; the block witnesses

supply $\alpha = 496$ without requiring a bound on the largest possible compatible subset of the Leech shell.

4.4. A two-layer configuration with 201567 points. The larger construction retains the first-layer heads of the public 201566-point contribution [16] and replaces its second layer by a 311-owner witness. We give the pair conditions explicitly. For this subsection use squared norm 32 and threshold 16 throughout. Write Σ_{27} for the Leech shell generated in the Golay coordinate basis of the two-layer data. All heads and owners below use that same basis. The shell argument in Appendix A applies to its verified Golay generator.

The first-layer data are 2258 distinct integer heads Y of squared norm 192, each assigned to one of four triples. In the order of the stored labels, these triples of roots are

$$\begin{array}{c|ccc} 0 & (1, 1, 0) & (-1, 0, -1) & (0, -1, 1) \\ 1 & (1, -1, 0) & (-1, 0, 1) & (0, 1, -1) \\ 2 & (-1, -1, 0) & (1, 0, -1) & (0, 1, 1) \\ 3 & (-1, 1, 0) & (1, 0, 1) & (0, -1, -1). \end{array}$$

For a head y in triple b , take the three points

$$(y/3, 4v/\sqrt{3}) \quad (v \text{ in triple } b). \quad (4.11)$$

The pure-tail points are $(0, 4Rv)$ for all twelve roots $v \in V$, with R as in (4.3).

A first-layer owner is a shell vector $z \in \Sigma_{27}$ with $y \cdot z > 48$. The finite data have 2090 heads with exactly one owner and 168 heads with none; all 2090 owners are distinct. The former are of the form $y = 3u + v$ with $u \in \Sigma_{27}$, $v^2 = 48$, and $u \cdot v = -24$. The latter are twice norm-48 vectors. These descriptions explain the candidate families; verification recovers the owners by checking the whole shell. For distinct first-layer heads, the stored data satisfy

$$y \cdot y' \leq \begin{cases} 48, & \text{on the same triple,} \\ 96, & \text{on different triples.} \end{cases} \quad (4.12)$$

Retaining all nonowners at the equator gives $196560 - 2090 + 3(2258) + 12 = 201256$ points.

The additional data are 311 distinct owners $u \in \Sigma_{27}$, with a label $\ell(u) \in \{1, 2, 3\}$. None is a first-layer owner. Replace the equatorial point $(u, 0)$ by

$$(\sqrt{3}u/2, \varepsilon\sqrt{8}e_{\ell(u)}), \quad \varepsilon \in \{-1, 1\}. \quad (4.13)$$

The three labels have respectively 105, 103, and 103 owners. The checked inequalities are

$$y \cdot u \leq 32 \quad \text{for every first-layer head } y, \quad (4.14)$$

$$u \cdot u' \leq 8 \quad \text{if } \ell(u) = \ell(u'), \quad u \neq u', \quad (4.15)$$

$$u \cdot u' \leq 16 \quad \text{if } \ell(u) \neq \ell(u'). \quad (4.16)$$

Theorem 4.5. *The two layers just specified, their retained equator, and their twelve pure-tail points form a kissing configuration of 201567 points in dimension 27.*

Proof. All points have squared norm 32. An old equatorial point and a first-layer cap have product at most $48/3 = 16$. The two thresholds in (4.12) give respectively $48/9 + 32/3 = 16$ and $96/9 + 16/3 = 16$. Two different caps of the same head have product $192/9 - 16/3 = 16$. The rotation estimate of Lemma 4.2, scaled by 32, checks the first layer against all pure tails.

A second-layer cap has product at most $(\sqrt{3}/2)16 = 8\sqrt{3} < 16$ with every retained equatorial point. Its own owner is deleted. Against a first-layer cap, the head product and tail product sum to at most

$$\frac{\sqrt{3}}{6}(32) + \frac{4\sqrt{8}}{\sqrt{3}} = \frac{16\sqrt{3} + 8\sqrt{6}}{3} < 16.$$

The last inequality is equivalent to $2\sqrt{3} + \sqrt{6} < 6$ and follows by squaring positive quantities. Second-layer caps on distinct owners of the same label have product at most $(3/4)8 + 8 = 14$; distinct labels give at most $(3/4)16 = 12$. The two caps of one owner have product $(3/4)32 - 8 = 16$. A second-layer cap and a pure tail have product at most $\sqrt{8}\sqrt{32} = 16$. All old-old pairs retain their original bounds.

The head squared norms in the equator, first layer, second layer, and pure-tail layer are respectively 32, $64/3$, 24, and zero, so different layers cannot coincide. Within a layer, distinct heads and tail labels give distinct points. Each of the 311 new owners is removed once and inserted twice. Hence

$$196560 - 2090 - 311 + 3(2258) + 2(311) + 12 = 201567.$$

Dividing all points by $\sqrt{32}$ proves the unit-sphere claim. $\square$

The finite optimization behind the second layer has a particularly small geometric description. Keep only shell vectors that are not first-layer owners and satisfy (4.14). Connect two such vectors when their product exceeds eight. One seeks a largest subset that can be split into three independent sets of this graph. Assign each independent set to one coordinate line. Distinct-label products are already controlled by the Leech-shell bound. The published 310-owner witness and the new 311-owner witness are feasible assignments in this same model; each extra owner contributes one net point.

5. LAYERED BINARY CODES AND JOINT SUPPORT EXCHANGES

The Leech lift gains points by changing how its finite blocks are used. The next construction instead keeps the layer geometry fixed and improves its binary ingredients. We first give the common geometric argument, then specify the support families and the dense sign codes. This order separates the dimension-independent proof from the finite objects that determine the cardinalities.

5.1. The three-layer construction. Let $n \geq m \geq 32$. Choose a binary code $T \subset \mathbb{F}_2^m$ with minimum distance at least $\lceil m/4 \rceil$, and a support family $\mathcal{B} \subseteq \binom{[n]}{8}$ such that

$$|B \cap B'| \leq 4 \quad (B \neq B', B, B' \in \mathcal{B}). \quad (5.1)$$

For the standard basis $e_0, \dots, e_{n-1}$ of $\mathbb{R}^n$, define

$$X_T = \left\{ \frac{1}{\sqrt{m}} \sum_{i=0}^{m-1} (-1)^{c_i} e_i : c \in T \right\}, \quad (5.2)$$

$$X_M = \left\{ \frac{1}{\sqrt{8}} \sum_{i \in B} \varepsilon_i e_i : B \in \mathcal{B}, \varepsilon_i \in \{\pm 1\}, \prod_{i \in B} \varepsilon_i = 1 \right\}, \quad (5.3)$$

$$X_R = \left\{ \frac{\varepsilon e_i + \varepsilon' e_j}{\sqrt{2}} : 0 \leq i < j < n, \varepsilon, \varepsilon' \in \{\pm 1\} \right\}. \quad (5.4)$$

The final layer consists of the normalized D_n roots.

Proposition 5.1 (Three-layer specialization of ERS [6]). *The union $X_T \cup X_M \cup X_R$ is a kissing configuration with*

$$|X| = |T| + 128|\mathcal{B}| + 2n(n-1). \quad (5.5)$$

Proof. All points are unit vectors. Their support sizes m , eight, and two distinguish the layers. Within each layer, the signs and support recover the parameters uniquely. On an eight-element support there are 2^7 even sign patterns; on a two-element support there are four sign patterns. This proves the count.

In the dense layer, two codewords at distance d have inner product $1 - 2d/m \leq 1/2$. On one middle support, two distinct even sign patterns differ in at least two positions, giving at most

$(8 - 4)/8 = 1/2$. On different middle supports, (5.1) gives at most four contributions of $1/8$. Two roots with different supports share at most one coordinate, giving at most $1/2$; on the same support distinct roots have inner product zero or -1 .

For the three cross-layer cases, the numbers of overlapping coordinates give respectively

$$\langle X_T, X_M \rangle \leq \sqrt{8/m}, \quad \langle X_T, X_R \rangle \leq \sqrt{2/m}, \quad \langle X_M, X_R \rangle \leq \frac{2}{\sqrt{8}\sqrt{2}} = \frac{1}{2}. \quad (5.6)$$

Here the notation denotes the maximum over the indicated two layers. The first two bounds are at most $1/2$ because $m \geq 32$. If a middle support extends beyond the first m coordinates, the overlap is smaller. Thus every pair satisfies (1.1). $\square$

The proof imposes no coupling between particular support words and particular sign words. Any inputs with the stated parameters can be combined. In particular, one additional support contributes 128 points, and an increase in the sign-code size contributes its full cardinality increment.

5.2. The support families. For eight-element supports the identity

$$d_H(B, B') = 16 - 2|B \cap B'|$$

identifies (5.1) with minimum distance eight. The following finite objects supply the middle layers.

Theorem 5.2. *The following constant-weight-code lower bounds hold:*

n	33	34	37	38	39
$A(n, 8, 8) \geq$	1803	1960	2843	3077	3383.

Finite verification. The complete binary-mask lists $\mathcal{B}_n$ are specified in Table 8. For the list of length n , every mask is nonnegative and less than 2^n , has exactly eight set bits, and occurs once. Every unordered pair has an intersection of size at most four. Direct pairwise checks give the distributions in Table 3. An independent equivalent check enumerates all five-element subsets of each support: no such subset occurs in two supports. This equivalence holds because an intersection has size at least five exactly when it contains a five-subset. In total $56|\mathcal{B}_n|$ distinct keys are obtained. These finite checks establish the stated codes. $\square$

TABLE 3. Support cardinalities and numbers h_j of unordered pairs with intersection size j . All other intersection counts are zero.

n	$ \mathcal{B}_n $	h_0	h_1	h_2	h_3	h_4
33	1803	134282	389920	680407	234225	185669
34	1960	168266	488949	783677	281186	197742
37	2843	443987	1255368	1461343	621750	257455
38	3077	555513	1521825	1676186	706467	272435
39	3383	718023	1896522	1985532	820689	299887

The final support families arise from exchanges in the constant-weight space. For an incumbent family $\mathcal{B}$ and a new support c , let $F(c) = \{b \in \mathcal{B} : |b \cap c| > 4\}$. If $Y \subseteq \binom{[n]}{8} \setminus \mathcal{B}$ is internally compatible, then

$$\mathcal{B}' = (\mathcal{B} \setminus F(Y)) \cup Y, \quad F(Y) = \bigcup_{c \in Y} F(c), \quad |\mathcal{B}'| - |\mathcal{B}| = |Y| - |F(Y)|. \quad (5.7)$$

The old-old and new-new pairs are compatible by assumption; all incompatible old-new pairs have been removed. The union in $F(Y)$ is essential: several inserted words may share deletion costs, allowing a positive joint exchange even when no single insertion improves the cardinality.

The final transitions are

n	33	34	37	38	39
parent size	1802	1959	2840	3076	3382
$ F(Y) $	9	6	37	6	14
$ Y $	10	7	40	7	15
final size	1803	1960	2843	3077	3383.

The certificate includes each complete parent and its inserted and removed sets. The checker recomputes each conflict union $F(Y)$ and reconstructs the final family by (5.7). The theorem itself uses the final complete witnesses, so its verification is independent of choices made during the search.

5.3. Dense layers from coset unions. Let $H \subset \mathbb{F}_2^m$ be linear and let $r_0, \dots, r_{q-1}$ be coset representatives. Define

$$T = \bigcup_{a=0}^{q-1} (r_a + H), \quad \delta_{ab} = \min_{h \in H} \text{wt}(r_a + r_b + h) \quad (a \neq b). \quad (5.8)$$

Within one coset, distances between distinct words are the nonzero weights of H . Between two cosets, the difference set is exactly $r_a + r_b + H$. Thus the δ_{ab} determine all cross-coset minimum distances. Positive δ_{ab} also prove that the cosets are disjoint, giving $|T| = q2^{\dim H}$.

Proposition 5.3. *The generators and representatives in Appendix B define the sign codes in Table 4.*

TABLE 4. Parameters of the dense sign codes.

m	$\dim H$	$d(H)$	q	$d(T)$	$ T $	Used in dimension
32	17	8	1	8	131072	32–37
38	14	12	11	10	180224	38
39	15	12	10	10	327680	39

Finite verification. Binary elimination gives the three displayed ranks. Enumeration of the spans gives the weight distributions in Appendix B, whose least positive weights are eight, twelve, and twelve. The length-32 code uses the sole representative zero. For lengths 38 and 39, enumeration of (5.8) gives $\delta_{ab} = 10$ for all 55 and 45 distinct representative pairs, respectively. The preceding difference-set argument proves the sizes and minimum distances. Finally, each distance is at least $\lceil m/4 \rceil$, as required by Proposition 5.1. $\square$

The length-32 code is the Cheng–Sloane [32, 17, 8] code [23], in the generator convention of Grassl’s table [31]. The other two kernels have a common quadratic-residue origin. Let

$$g(x) = x^{23} + x^{19} + x^{18} + x^{14} + x^{13} + x^{12} + x^{10} + x^9 \\ + x^7 + x^6 + x^5 + x^3 + x^2 + x + 1 \in \mathbb{F}_2[x]. \quad (5.9)$$

Take the coefficient vectors of $x^i g(x)$, $0 \leq i < 24$, as length-47 binary words, and append their parity at coordinate 47. This specifies the extended quadratic-residue code in the chosen convention [3, Chapter 16]. Shorten at coordinates 40, $\dots$, 47, retaining coordinates 0, $\dots$, 39 in order, to obtain a dimension-16 code H_{40} . Shortening H_{40} at coordinate zero gives the printed H_{39} ; shortening at coordinates zero and one gives the printed H_{38} . Here shortening imposes zeros before

deleting coordinates. Exact row-space comparisons verify these identities and, in the displayed retained-coordinate conventions,

$$H_{38} = \{(h_0, \dots, h_{37}) : (h_0, \dots, h_{37}, 0) \in H_{39}\}. \quad (5.10)$$

These reconstructions give both kernels a precise common source. The distance proof uses their explicit generators and the affine representatives rather than an assumed distance of a named mother code.

The five layered-code constructions. For each $n \in \{33, 34, 37, 38, 39\}$, combine the support family of Theorem 5.2 with the sign code of Proposition 5.3. Proposition 5.1 gives Table 5. The dimension-37 and dimension-38 counts are improved further in Sections 6 and 7. $\square$

TABLE 5. The three contributions to the five code-based configurations.

n	$ X_T $	$ X_M $	$ X_R $	Total
33	131072	230784	2112	363968
34	131072	250880	2244	384196
37	131072	363904	2664	497640
38	180224	393856	2812	576892
39	327680	433024	2964	763668

At the first three lengths, the increase relative to the cited comparison is 128 times the increase in support size, respectively 15, 24, and 11. At the last two lengths, both the larger support families and the explicit coset unions enter the final count. The same geometric argument accommodates all five inputs, while their different dense-layer distances lead to the different local geometries considered next.

5.4. A new length-32 support family. The same construction applies to a verified family of 1676 eight-subsets of $[32]$. There are 1,403,650 unordered support pairs. Their intersection histogram, in orders zero through four, is

$$(112077, 318935, 601033, 196122, 175483).$$

No larger intersection occurs. Together with the same $[32, 17, 8]$ dense code, Proposition 5.1 gives

$$K(32) \geq 2^{17} + 128(1676) + 2(32)(31) = 347584.$$

The family was obtained from Echols's public 1667-word seed [18] through joint exchanges. The final witness, rather than successful replay of a randomized search, is the finite input to this proof. The 2843-word length-37 family above is also improved to 2845 words. Section 6 uses this larger family and changes the sign patterns, giving a further gain.

6. LOCAL SIGNED REPLACEMENTS

6.1. A compatible replacement region. Use the three layers of Proposition 5.1, scaled to squared norm 32, with the same binary $[32, 17, 8]$ code. Its 2^{17} codewords give vectors $(\pm 1)^{32}$, padded with zeros to dimension d . Let $\mathcal{B}$ be a family of eight-subsets of $\{0, \dots, d-1\}$ with pairwise intersections at most four. For each support take the 128 sign patterns of a fixed parity, with nonzero entries ± 2 . Finally take all vectors with two entries ± 4 . These are the standard ERS layers [6], all of squared norm 32.

For $d = 35$ and 36 , the archived support families have 2,157 and 2,742 blocks, respectively [17]. The base cardinalities are

$$2^{17} + 128 \cdot 2157 + 4 \binom{35}{2} = 409548, \quad 2^{17} + 128 \cdot 2742 + 4 \binom{36}{2} = 484568.$$

Lemma 6.1 (Isolated replacement). *Let W be a coordinate set containing exactly b blocks of $\mathcal{B}$. Suppose every other block meets W in at most four coordinates. Let $Q \subset \{0, \pm 1\}^W$ consist of distinct weight-eight vectors with $q \cdot q' \leq 4$ for distinct q, q' . Replacing the $128b$ middle-layer vectors supported inside W by $2Q$ preserves the kissing condition and changes the cardinality by $|Q| - 128b$.*

Proof. Two new vectors have inner product at most 16. A new vector and any retained middle-layer vector overlap in at most four coordinates, so their inner product is also at most 16. A top-layer vector has inner product at most $8 \cdot 2 = 16$ with a new vector. A bottom-layer vector has inner product at most $2 \cdot 4 \cdot 2 = 16$. All new vectors have squared norm 32 and are distinct from the retained layers. The unchanged layers retain their original compatibility. $\square$

In dimension 35 take $W = \{12, \dots, 22\}$; precisely block 2145 lies inside W . In dimension 36 take $W = \{12, \dots, 23\}$; precisely blocks 2739, 2740 and 2741 lie inside W . Indices are zero-based. The independent checker verifies these facts from the complete support lists, together with all boundary intersections.

6.2. Hadamard transfer.

Lemma 6.2 (Paired Hadamard transfer). *Let $\mathcal{A}$ be a family of four-subsets on an even number m of coordinates, with pairwise intersections at most two. Fix a perfect matching of the coordinates and retain the blocks meeting each matched pair in at most one coordinate. If r blocks survive, there exists a signed weight-eight code of size $16r$ in $\{0, \pm 1\}^m$ whose distinct vectors have inner products at most four.*

Proof. Take all sixteen sign patterns on each retained block and apply $H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ to each matched coordinate pair. Each occupied pair contains exactly one input entry ± 1 , and its image contains two entries ± 1 . Thus every image has weight eight. Since $H^T H = 2I$, all inner products double. For different supports the original inner product is at most two. For the same support, two distinct sign patterns differ in at least one position, giving inner product at most $4 - 2 = 2$. The map is invertible and preserves distinctness. $\square$

For dimension 35, an exact-cover construction gives a Steiner quadruple system $S(3, 4, 10)$ with 30 blocks. The matching

$$(0, 1), (2, 3), (4, 5), (6, 7), (8, 9)$$

retains 16 blocks, hence 256 signed weight-eight vectors. One extra zero coordinate embeds them into the eleven-coordinate region W .

The twelve-coordinate source is the 51-block design of Kalbfleisch–Stanton [26, Theorem 7], also used in the local-replacement construction of Takhanov–Yun [25, Section V]. For dimension 36, use two copies of a one-factorization of K_6 . In each six-coordinate half, take the complements of the three edges in one factor, giving three internal four-subsets. For each of the five factors, combine each of its three edges in the first half with each of the three in the second half. There are $3 + 3 + 5 \cdot 9 = 51$ blocks, with intersections at most two. The matching

$$(0, 1), (2, 4), (3, 5), (6, 7), (8, 10), (9, 11)$$

retains 36 blocks, hence 576 signed weight-eight vectors. The two matched designs supply the replacement codes for the isolated ERS regions described above.

Proposition 6.3 (A four-colour construction). *There is a signed weight-eight code of size 576 in twelve coordinates with distinct-pair inner products at most four.*

Proof. Use the edges of one colour of the one-factorization as the coordinate pairs in each half. Each of the other four colours has three edges transverse to these pairs. For each colour choose one edge in each half and take their union. This gives $4 \cdot 3 \cdot 3 = 36$ transverse four-subsets. Their intersections are at most two: within a colour they can share one edge; between colours they share at most one endpoint per half. Lemma 6.2 supplies $16 \cdot 36 = 576$ signed vectors. $\square$

Proposition 6.4 (Sign-extension bound). *Let $Q \subset \{0, \pm 1\}^m$ have constant weight w and distinct-pair inner products at most s . If $2w - m > s$, then $|Q| \leq 2^w$. In particular, the maximum size of a weight-eight code in eleven coordinates with inner products at most four is 256.*

Proof. A full sign string $\sigma \in \{\pm 1\}^m$ cannot extend two members of Q : their supports would intersect in at least $2w - m > s$ coordinates, and their signs agree wherever they intersect. Each member has 2^{m-w} extensions. Hence $2^{m-w}|Q| \leq 2^m$. The 256-point transferred code above attains the bound for $(m, w, s) = (11, 8, 4)$. $\square$

Theorem 6.5. *There are kissing configurations of sizes 409,676 in dimension 35 and 484,760 in dimension 36.*

Proof. The two replacement sizes give

$$N_{35} = 2^{17} + 128(2157 - 1) + 256 + 4 \binom{35}{2} = 409676,$$

$$N_{36} = 2^{17} + 128(2742 - 3) + 576 + 4 \binom{36}{2} = 484760.$$

The preceding lemmas prove compatibility involving the new vectors. For completeness, the top layer has inner product at most $32 - 2 \cdot 8 = 16$. On one retained middle support, equal parity forces at least two sign flips, so the inner product is at most $32 - 8 \cdot 2 = 16$; different supports intersect in at most four positions. The bottom layer has maximum inner product 16. The remaining cross-layer bounds are 16 for top–middle, 8 for top–bottom, and 16 for middle–bottom. All layers have squared norm 32. The archived finite data establish every hypothesis of these arguments. $\square$

6.3. A 512-point replacement in dimension 37. The replacement argument only needs every new support to intersect every retained support in at most four coordinates. It need not impose this condition on an entire containing coordinate region. Take the verified 2845-word length-37 family and delete three complete sign bundles. The replacement consists of 512 signed weight-eight vectors on twelve coordinates. In the source’s one-based, left-to-right numbering, the paired coordinates are

$$(18, 20), (23, 24), (15, 21), (16, 22), (14, 17), (19, 25).$$

Thirty-two transverse four-subsets on the twelve input axes, each with all sixteen signs, give the 512 vectors by the preceding Hadamard transfer. The four-subsets, coordinate pairs and signed images are supplied in `constructions/codes/data/d37_signed_patch.json`. The checker regenerates all $32 \cdot 16 = 512$ images and verifies equality with the supplied vector set. Their internal products are at most four, and every image support meets every retained support in at most four coordinates. Hence

$$K(37) \geq 2^{17} + 128(2845 - 3) + 512 + 2(37)(36) = 498024.$$

Against the 2832-support comparison family, the thirteen extra supports contribute $13 \cdot 128 = 1664$ points. The signed replacement contributes $512 - 3 \cdot 128 = 128$ further points, for a total gain of 1792.

The left-to-right numbering is part of the witness specification. Assigning coordinate one to the rightmost bit instead requires the reversal $i \mapsto 38 - i$ for $1 \leq i \leq 37$. The independent verifier checks the complete support and signed-vector conditions in one consistent convention. Equivalently, left-to-right coordinate i is integer bit $37 - i$ when the least significant bit has index zero.

7. SECOND-SHELL COMPLETION IN DIMENSION 38

The preceding layered-code construction gives 576892 points in dimension 38. The larger bound in Table 1 starts from the public 591612-point configuration [15]. Its decomposition has an unoccupied equator. This permits the addition of directions from a different Leech shell.

Lemma 7.1 (Equatorial completion). *Suppose $X \subset S^{D+k-1}$ is a kissing configuration consisting of points $(u/\sqrt{q}, t)$ and pure-tail points $(0, p)$. Let $V \subset \mathbb{R}^D$ be a set of representatives of distinct antipodal lines, all of squared norm q , such that*

$$|\langle v, v' \rangle| \leq q/2 \quad (v, v' \in V, v \neq v'), \quad (7.1)$$

$$|\langle v, u \rangle| \leq q/2 \quad (v \in V, (u/\sqrt{q}, t) \in X). \quad (7.2)$$

Assume every t in a point with nonzero head is nonzero. Then adjoining all $(\pm v/\sqrt{q}, 0)$ gives a kissing configuration with $|X| + 2|V|$ points.

Proof. New points are unit vectors and their products with old pure-tail points vanish. Their products with old points having a head equal $\pm \langle v, u \rangle/q$, so (7.2) gives the required bound. Products between different new lines are bounded by (7.1); the opposite points on one line have product -1 . The nonzero-tail assumption and distinctness of the new lines prove that the added points are all new. $\square$

7.1. The base configuration. Use the integer Golay–Leech shell Σ of squared norm 32, with the Golay generator stored in the dimension-38 package. Partition its 98280 antipodal lines into classes with pairwise absolute inner products at most eight. The tail data consist of 1932 vectors of squared norm eight in $\mathbb{R}^{14}$, partitioned into 644 zero-sum triples. Inside a triple the mutual products are -4 ; between any distinct tail vectors they are at most four.

Each mother line is assigned a triple. It contributes six points of the form

$$(\varepsilon u/\sqrt{48}, d/\sqrt{24}), \quad \varepsilon \in \{-1, 1\}, \quad (7.3)$$

where d runs through its assigned triple. Their squared norm is $32/48 + 8/24 = 1$. Pairs in one class at one tail have product at most $8/48 + 8/24 = 1/2$. Repeated heads at different tails in their triple have product $32/48 - 4/24 = 1/2$. Different heads and different tails have product at most $16/48 + 4/24 = 1/2$. The cases with an opposite head have a smaller head contribution or are antipodal.

There are also 1932 pure-tail points. Write the tail vectors as the rows of an integer matrix F , and the stored rational rotation as M/h , with M integral and $h > 0$. Its defining checks are

$$MM^t = h^2 I_{14}, \quad |(FM^t F^t)_{ij}|^2 \leq 48h^2 \quad \text{for all } i, j. \quad (7.4)$$

The pure-tail points are $(0, dM^t/(h\sqrt{8}))$. Orthogonality preserves their norms and mutual products. Their products with the caps in (7.3) are entries of $FM^t F^t/(h\sqrt{192})$, so the second inequality gives absolute products at most $1/2$. Thus both required checks use exact integer arithmetic. The layer counts are

$$\underbrace{0}_{\text{equator}} + \underbrace{6(98280)}_{\text{caps}} + \underbrace{1932}_{\text{pure tails}} = 591612.$$

The class partition, triples, rotation, and Golay generator are inherited data from the public construction; the completion below is the additional part.

7.2. The 144 added lines. The use of another shell removes the old shell from the selection constraints through a general minimum-norm argument.

Lemma 7.2 (Cross-shell compatibility). *Let $\mathcal{L}$ be a lattice of minimum squared norm m . If $u, v \in \mathcal{L}$ have distinct squared norms a, b , then $|u \cdot v| \leq (a + b - m)/2$. In the integer Leech model, $a = 32$, $b = 48$ and $m = 32$ give $|u \cdot v| \leq 24$.*

Proof. The distinct norms imply $u \pm v \neq 0$. Applying the minimum bound to these two lattice vectors gives $a + b \pm 2u \cdot v \geq m$. $\square$

The new certificate lists 144 integer rows v of squared norm 48. They are distinct modulo sign and satisfy

$$|v \cdot v'| \leq 24 \quad (v \neq v'), \quad \max_{u \in \Sigma} |v \cdot u| \leq 24. \quad (7.5)$$

Each row also belongs to the Leech lattice. In the Golay convention used for these arrays, membership means that for some $\varepsilon \in \{0, 1\}$ all coordinates have parity ε , $((v_i - \varepsilon)/2 \bmod 2)_i$ is a Golay word, and $\sum_i v_i \equiv 4\varepsilon \pmod{8}$. The cross-shell inequality therefore follows from Lemma 7.2; the mutual products are the remaining selection constraint. The checker verifies these congruences and also recomputes every cross product as an independent finite check. The shell Σ is generated from the same Golay basis as the base construction; a change of coordinate convention must be applied to both arrays.

Theorem 7.3. *There is a kissing configuration of 591900 points in dimension 38.*

Proof. Apply Lemma 7.1 with $q = 48$ to the base configuration (7.3) and the 144-line witness. Its hypotheses are exactly (7.5). The new points have zero tail, while every old cap has tail norm $1/\sqrt{3}$; the old pure-tail points have zero head. Thus the count is $591612 + 2(144) = 591900$. $\square$

The completion changes the shell used for new equatorial points. All minimal directions have already been used in the cap layer, but that does not exhaust the admissible directions of the equatorial sphere. The norm-48 shell supplies a separate finite search space with compatible cap products. Searching in that space produced a 142-line witness and then the 144-line witness. The latter, including its complete base checks, is the input to Theorem 7.3. The second-shell choice is useful because it has the normalization of the cap heads and its entire interaction with the first shell is controlled by the lattice minimum.

8. SEVEN LIFTS FROM A COMPATIBLE LINE CLASS

The antipodal block lift amplifies improvements to a single compatible class. Two choices determine the gain: the size of the class and the overlap among its selected images. This section makes both choices explicit for P_{48p} and obtains seven configurations from one geometric argument.

8.1. The tail configurations. For dimensions $k = 1, \dots, 7$, use respectively the normalized root configurations $A_1, A_2, D_3, D_4, D_5, E_6, E_7$. They have 2, 6, 12, 24, 40, 72, 126 points and are antipodal. Explicitly, $A_1 = \{-1, 1\}$, and A_2 is the six points $(e_i - e_j)/\sqrt{2}$ in the plane $x_1 + x_2 + x_3 = 0$. The D_k roots are $(\varepsilon e_i + \varepsilon' e_j)/\sqrt{2}$, with $i < j$. Their kissing conditions follow directly from their coordinate overlaps.

For completeness the last two codes can be specified together in an integer model. Let

$$W = \{\pm 2e_i \pm 2e_j : i < j\} \cup \{(\varepsilon_0, \dots, \varepsilon_7) : \varepsilon_i \in \{\pm 1\}, \prod_i \varepsilon_i = 1\}. \quad (8.1)$$

Every row has squared norm eight and distinct rows have inner product at most four: this follows from support overlap for the first family, at most two signed contributions for cross-family pairs, and an even positive sign distance within the second family. The subsets

$$W_7 = \{w \in W : w_6 + w_7 = 0\}, \quad W_6 = \{w \in W : w_5 = w_6 = -w_7\}$$

have respectively $62 + 64 = 126$ and $40 + 32 = 72$ rows. The maps

$$w \mapsto (w_0, \dots, w_5, \sqrt{2}w_6), \quad w \mapsto (w_0, \dots, w_4, \sqrt{3}w_5)$$

are isometries on the indicated subspaces. Dividing by $\sqrt{8}$ therefore gives the required 7- and 6-dimensional antipodal codes.

8.2. The 7077-line class and its automorphism images. We use the catalogue lattice P_{48p} in its published integral Gram basis G [29]. Its minimum squared norm is six. Since it is even unimodular of rank 48, the theta-series calculation in (C.6) gives 52416000 minimal vectors. The witnesses use this catalogue basis directly. The quadratic-residue neighbour used in dimension 45 is constructed in Section 10.1. The theta-series calculation in Appendix C applies to every extremal even unimodular lattice of rank 48.

Represent a minimal line by an integer row c satisfying $cGc^t = 6$, identifying c and $-c$. The new mother class $\mathcal{C}$ has 7077 lines and satisfies

$$|cGc^t| \leq 2 \quad (c \neq \pm c', \ c, c' \in \mathcal{C}). \quad (8.2)$$

Every norm and pair product in this assertion is an integer. The certificate supplies the complete class and matrices A_i with $A_iGA_i^t = G$. Thus $\mathcal{C}A_i$ is another class with the same property.

For t selected images, assign each line to the first image in which it occurs. Write $\mathcal{D}_i$ for the assigned subset of $\mathcal{C}A_i$, and put

$$S = \left| \bigcup_{i=1}^t \mathcal{C}A_i \right| = \sum_{i=1}^t |\mathcal{D}_i|.$$

The $\mathcal{D}_i$ are disjoint as line sets. Taking both signs of each assigned line gives disjoint vector blocks B_i of size $2|\mathcal{D}_i|$, with normalized inner products at most $1/3$ by (8.2). Assign these blocks injectively to the t distinct antipodal lines of the corresponding root tail code. Theorem 2.5 gives

$$K(48 + k) \geq 52416000 + 2S + \tau_k, \quad \tau_k = 2t. \quad (8.3)$$

The injective assignment is part of the construction. For two different assigned classes, the minimal-norm bound gives a normalized head product at most $1/2$, and distinct tail lines give a tail product at most $1/2$. Their cap product is therefore at most $(3/4)(1/2) + (1/4)(1/2) = 1/2$. Reusing a tail line for arbitrary different classes would instead give $5/8$ and is not permitted.

Theorem 8.1. *The class and image matrices in the supplementary data give the seven bounds in Table 6.*

Proof. The finite checker verifies the catalogue Gram matrix, its exact isometries, all 7077 norms, distinctness modulo sign, and all distinct-line inequalities (8.2). It then reconstructs each selected image and counts its union with the preceding images. This gives the S values in the table. The root tail codes have the ranks, sizes, and inner products established above. Hence (8.3) applies. $\square$

8.3. Class growth and image selection. The public comparison uses a 7069-line class $\mathcal{C}_0 \subset \mathcal{C}$ [15]. With the selected matrices held fixed, let

$$S_0 = \left| \bigcup_{i=1}^t \mathcal{C}_0A_i \right|,$$

and let S_{ref} be the union size in the comparison construction. The increase in the final point count splits exactly as

$$2(S - S_{\text{ref}}) = 2(S - S_0) + 2(S_0 - S_{\text{ref}}). \quad (8.4)$$

TABLE 6. The line-class liftings. The union, rather than the sum of image sizes, determines the number of available lines.

d	t	S	τ_k	$52416000 + 2S + \tau_k$
49	1	7077	2	52430156
50	3	21231	6	52458468
51	6	42459	12	52500930
52	12	84874	24	52585772
53	20	141328	40	52698696
54	36	253851	72	52923774
55	63	442507	126	53301140

The first term measures the effect of enlarging the class with these images fixed; the second measures the effect of changing the images with the original class fixed. Both are obtained by exact union counts.

TABLE 7. The two contributions to the seven point-count improvements.

d	S_0	$2(S - S_0)$	$2(S_0 - S_{\text{ref}})$	Total
49	7069	16	0	16
50	21207	48	2	50
51	42411	96	18	114
52	84778	192	64	256
53	141168	320	154	474
54	253565	572	296	868
55	442009	996	414	1410

The three selected images in dimension 50 are disjoint. Larger image families overlap, so an eight-line enlargement need not supply eight new lines per image. In dimension 55 it contributes 498 distinct lines, rather than $8 \cdot 63 = 504$. The image choice contributes a further 207 lines. Thus the two improvements reinforce one another through the union, rather than through a sum of independently optimized class sizes.

9. EMBEDDED ROOT SYSTEMS AND MOMENT-CERTIFIED SECTIONS

An embedded root system imposes restrictions on the projections of the entire minimal shell. In the case below, these restrictions determine the number of points perpendicular to a child section.

Theorem 9.1 (E_8 -extension count). *Let L be an extremal even unimodular lattice of rank 48, and let $\iota : \sqrt{3}E_8 \hookrightarrow L$ be an isometric embedding. Use the simple-root numbering in (9.5), and set*

$$U = \text{span}_{\mathbb{R}}\{\iota(\sqrt{3}\alpha_1), \dots, \iota(\sqrt{3}\alpha_5)\}.$$

Then exactly 2,553,792 vectors of squared norm 6 in L are perpendicular to U . The same count holds for every $W(E_8)$ -conjugate of this D_5 subsystem inside the embedded E_8 .

The proof uses integrality, the lattice minimum and spherical-design moments. It does not require the Weyl group to extend to automorphisms of L . We first give an explicit realization, then establish the count under the hypotheses of the theorem.

9.1. A realization in P_{48n} . Take $L = P_{48n}$, whose Gram matrix G is published as CQ48a in the Nebe–Sloane catalogue [28]. It is an extremal even unimodular lattice of rank 48, minimum squared norm 6, with minimal shell

$$X = \{x \in L : \|x\|^2 = 6\}, \quad |X| = 52,416,000.$$

The parent certificate [19] specifies eight integer coefficient rows T and scaled coordinate rows B , satisfying

$$TGT^t = H = 3C(E_8) = BB^t/96. \quad (9.1)$$

Thus the rows of T , interpreted in the basis of L , generate an actual $\sqrt{3}E_8$ sublattice. The catalogue Gram and the certificate Gram agree entry by entry. Let T_5 consist of the first five rows of T . Its Gram is

$$H_5 = \begin{pmatrix} 6 & 0 & -3 & 0 & 0 \\ 0 & 6 & 0 & -3 & 0 \\ -3 & 0 & 6 & -3 & 0 \\ 0 & -3 & -3 & 6 & -3 \\ 0 & 0 & 0 & -3 & 6 \end{pmatrix}, \quad \det H_5 = 972. \quad (9.2)$$

This is $3C(D_5)$ in the stated numbering. Both T and T_5 have all nonzero Smith invariants equal to one. In particular the embedded sublattices are primitive. If U is the span of the five rows, then $\dim U^\perp = 43$.

Corollary 9.2. *The set $Y = X \cap U^\perp$ has 2,553,792 elements. Consequently the ordinary Euclidean kissing number satisfies $K(43) \geq 2,553,792$.*

The cardinality follows from Theorem 9.1 and the verified embedding (9.1). To verify the kissing condition, distinct $x, x' \in Y$ satisfy $x - x' \in L \setminus \{0\}$, so $12 - 2\langle x, x' \rangle \geq 6$. Hence $Y/\sqrt{6}$ consists of unit vectors with pairwise inner products at most $1/2$.

The construction is specified without a list of millions of points. In integer basis coordinates its defining set is

$$\mathcal{Z} = \{z \in \mathbb{Z}^{48} : zGz^t = 6, T_5Gz^t = 0\}. \quad (9.3)$$

For any real E with $EE^t = G$, the points are $zE/\sqrt{6}$, for $z \in \mathcal{Z}$. All matrices needed for this definition are included in the accompanying source certificate.

9.2. The complete projection domain. Now let L be any lattice satisfying Theorem 9.1, and let X be its squared-norm-six shell. The theta-series identity (C.6) gives $|X| = 52,416,000$. Identify the parent section with $\sqrt{3}E_8$, with simple roots $\sqrt{3}\alpha_1, \dots, \sqrt{3}\alpha_8$. Write the orthogonal projection of $x \in X$ onto this section as $p = \lambda/\sqrt{3}$. Integrality of L and self-duality of E_8 imply $\lambda \in E_8$. Set

$$y_i = \langle \lambda, \alpha_i \rangle = \langle x, \sqrt{3}\alpha_i \rangle, \quad q = \|p\|^2 = \|\lambda\|^2/3 = y^t H^{-1} y \leq 6.$$

For an E_8 root r , if $x \neq \pm\sqrt{3}r$, applying the minimum bound to $x \pm \sqrt{3}r$ gives

$$|\langle \lambda, r \rangle| \leq 3. \quad (9.4)$$

The exceptions are $\lambda = 3r$. Each such projection has multiplicity one: its squared length is already 6, so the perpendicular component vanishes.

For a transparent finite enumeration use the model

$$E_8 = \{u/2 : u \in \mathbb{Z}^8, u_i \text{ all of the same parity, } \sum_i u_i \equiv 0 \pmod{4}\}.$$

Its roots are the 112 vectors $\pm e_i \pm e_j$ and the 128 half-integer vectors with entries $\pm 1/2$ and an even number of minus signs. The simple roots used here are

$$\begin{aligned} \alpha_1 &= (1, -1, -1, -1, -1, -1, -1, 1)/2, & \alpha_2 &= e_1 + e_2, \\ \alpha_{i+2} &= -e_i + e_{i+1} & & (1 \leq i \leq 6). \end{aligned} \quad (9.5)$$

Their Gram is precisely $H/3$.

Enumeration of $\|u\|^2 \leq 72$ subject to (9.4) gives

$\ \lambda\ ^2$	0	2	4	6	8
Nonexceptional signatures	1	240	2,160	6,720	17,280

There are no other nonexceptional signatures. Including the 240 exceptions gives 26,641 signatures in total.

The enumeration has an elementary completeness bound. Pair roots imply that the two largest absolute coordinates of u sum to at most 6. For the half-integer roots, the largest signed sum with an even number of minus signs is $\|u\|_1$, except when all coordinates are nonzero and the number of negative coordinates is odd; then it is $\|u\|_1 - 2 \min_i |u_i|$. This quantity must be at most 12. Since $\min_i |u_i| \leq 3$, the safe additional bound is $\|u\|_1 \leq 18$. The verifier exhausts these integer boxes and applies the parity, norm and root bounds exactly. The norm-eight shell loses the 240 doubled roots, leaving $17,520 - 240 = 17,280$ admissible points.

9.3. Moments and the averaging explanation. Venkov's theorem, as stated in Nebe [32, Theorem 4.1], makes X a spherical 11-design. For $2m \leq 10$ and section vectors $b_{i_1}, \dots, b_{i_{2m}}$, its moments therefore satisfy

$$\sum_{x \in X} \prod_{j=1}^{2m} \langle x, b_{i_j} \rangle = \frac{|X|6^m}{48 \cdot 50 \cdots (48 + 2m - 2)} \sum_P \prod_{(a,b) \in P} H_{i_a i_b}, \quad (9.6)$$

where P runs over pairings of the $2m$ labelled positions. The empty product gives the degree-zero equation.

Average the actual projection multiplicities under the abstract Weyl group $W(E_8)$. The nonexceptional domain has one orbit for each of its five norms. Let their sizes be s_j , projected squared norms be $q_j = 0, 2/3, 4/3, 2, 8/3$, and averaged multiplicities be a_j . Spherical radial moments give

$$\sum_j s_j a_j q_j^k = |X| 6^k \frac{(4)_k}{(24)_k} - 240 6^k, \quad 0 \leq k \leq 5, \quad (9.7)$$

where $(a)_k$ denotes the rising factorial. The ratio is the radial moment of an eight-dimensional orthogonal projection of a sphere in dimension 48. The subtraction removes the actual section roots. The equations for $k = 0, \dots, 4$ yield

$$(a_0, a_1, a_2, a_3, a_4) = (1, 092, 000, 116, 235, 9, 180, 495, 63/4).$$

The equation for $k = 5$ is an additional exact check.

The roots perpendicular to the first five simple roots form A_3 . The target subspace contains respectively 1, 12, 6, 24, 0 admissible signatures at the five norms, and 12 exceptional roots. Thus the average contact count over the Weyl images of this D_5 child is

$$1, 092, 000 + 12(116, 235) + 6(9, 180) + 24(495) + 12 = 2, 553, 792. \quad (9.8)$$

Every such image is an actual child inside the embedded E_8 . At least one child has a count at least as large as the average, proving the lower bound. The averaging takes place within the section's Weyl group; its transformations need not extend to automorphisms of L . The a_j are orbit averages of integer fibre multiplicities and can therefore be rational, as the value $63/4$ illustrates.

9.4. Exact count for the specified child. To prove the stronger cardinality claim for the first five rows, average only under the subgroup

$$\Gamma = \langle W(D_5), W(A_3), -1 \rangle.$$

This group preserves both the full admissible domain and the target $y_1 = \cdots = y_5 = 0$. Consequently averaging preserves the target count, even when the actual multiplicity function has no such symmetry. The invariant spherical moment equations remain valid after averaging.

The domain has 43 Γ -orbits O_j . Write v_j for the averaged multiplicity of a signature in O_j . A monomial moment row has entries

$$A_{\alpha,j} = \sum_{y \in O_j} y^\alpha,$$

and right-hand side b_α given by (9.6). Actual-root orbits supply the additional equations $v_j = 1$. The target is

$$c_j = |\{y \in O_j : y_1 = \cdots = y_5 = 0\}|.$$

The supplied certificate selects 43 moment or fixed-root rows, and provides a rational column vector w such that

$$A^t w = c, \quad w^t b = 2,553,792. \quad (9.9)$$

Since the actual averaged distribution satisfies $Av = b$, these identities give $|X \cap U^\perp| = c^t v = w^t Av = w^t b$. The domain, the matrices A, c , and the moment vector b depend only on the specified root system and the rank-48 spherical moments. Thus the identity holds for every L in Theorem 9.1. Precomposing ι with any $g \in W(E_8)$ proves the assertion for the conjugate subsystems, without extending g to L .

The full arrays, row labels, orbit representatives and rational w form the target-count certificate listed in Table 8. The independent verifier reconstructs all orbit entries, all entries of A , all moment right-hand sides b , and the target c from the standard coordinate model. Only the row selection and candidate w are taken from the certificate before checking (9.9). Thus verification does not reduce to trusting stored matrix entries.

9.5. What accounts for the improvement. The comparison certificate from [19] records 2,545,056 contacts. Its section is also a primitive $\sqrt{3}D_5$, with determinant 972, singleton cluster and parameter $\beta = 0$. The old and new section Grams are permutation-isometric. The improvement therefore concerns the embedding of the section, not its intrinsic lattice type or determinant.

The new section is a child of a specified E_8 inside P_{48n} ; the comparison uses a D_5 in P_{48p} . The parent E_8 supplies additional roots and projection relations that force the larger perpendicular shell population. Two isometric sections can have different numbers of ambient minimal vectors perpendicular to them. Their extension structure is therefore relevant to contact-number optimization even when determinant and singleton packing density agree.

10. A MOMENT-CERTIFIED PROJECTION IN DIMENSION 45

We obtain 7380090 points by projecting three fibres of the minimal shell of a quadratic-residue neighbour. The argument first establishes the minimum of this lattice, then bounds the three fibre sizes by an exact moment inequality and 298 explicit vectors.

10.1. The specified quadratic-residue neighbour. Let C be the extended ternary quadratic-residue code of length 48. We use its published parameters [48, 24, 15] [33, 34]. The supplement constructs a generator by a degree-23 factor of $1 + x + \cdots + x^{46}$ over $\mathbb{F}_3$ and a self-dual extension coordinate. The finite generator matrix fixes the factor, coordinate order and extension. Set

$$M = \{z \in \mathbb{Z}^{48} : z \bmod 3 \in C\}, \quad L = M/\sqrt{3}, \quad L_0 = \{z/\sqrt{3} \in L : z^2 \equiv 0 \pmod{6}\}.$$

An integer basis B_0 for M and a characteristic numerator $\chi \in M$ are supplied. They satisfy

$$\chi_i \text{ odd for all } i, \quad \chi^2 = 264, \quad \chi \cdot z \equiv z^2 \pmod{6} \quad (z \in M).$$

The last congruence is checked on B_0 and extends by integrality. Define

$$\Lambda = L_0 \cup (\chi/(2\sqrt{3}) + L_0).$$

The characteristic relation and $\chi^2/12 = 22$ make Λ even and integral. The two index-two constructions give covolume one. The supplied integer basis B satisfies $BB^t = 12G$ for the exact Gram matrix G of the 298-vector certificate. It is reconstructed from B_0 and the two cosets by integer row operations; the verifier checks equality with the archived G , not merely its determinant.

Lemma 10.1. *The specified lattice Λ has minimum squared norm six.*

Proof. In L_0 , a putative vector of squared norm two or four has numerator z with $z^2 \leq 12$. The minimum code weight forces $z \bmod 3 = 0$. Its nonzero squared norm is then $3w^2$, whose positive even values are at least six.

In the other coset, the numerator $y = \chi + 2z$ has all coordinates odd, so $y^2/12 \geq 4$. Equality requires $y_i = \pm 1$. Let $s_i \in \{-1, 1\}$ represent $\chi_i \bmod 3$. The certificate checks that $s \in C$ and $\chi \cdot s = 66$. Multiplying coordinates by s yields an equivalent extremal ternary code containing the all-one word. Its binary-valued words have weights 0, 24, 48, by the two-symbol specialization of the complete weight enumerator [35, equation (20)]. Write $y = s(1 - 2b)$ coordinatewise. Since $y \bmod 3 \in C$, the binary word b belongs to the normalized code, and its weight w is one of these three values. But $\chi_i s_i \equiv 1 \pmod{6}$, so

$$\chi \cdot z = \frac{\chi \cdot s - \chi^2}{2} - \sum_i \chi_i s_i b_i \equiv -99 - w \equiv 3 \pmod{6}.$$

This contradicts $z/\sqrt{3} \in L_0$. Evenness excludes other positive squared norms below six, and $\sqrt{3}(e_0 + e_1) \in L_0$ has squared norm six. $\square$

The weight-24 theta-series calculation in (C.6) now gives a minimal shell X of size 52416000. It is a spherical 11-design [32]. These are consequences for this specified Gram matrix, with the literature inputs and basis identities stated above.

10.2. A complete signature domain and its moments. The three stored section rows T satisfy

$$H = TGT^t = 8I_3 - 2J_3 = \begin{pmatrix} 6 & -2 & -2 \\ -2 & 6 & -2 \\ -2 & -2 & 6 \end{pmatrix}, \quad H^{-1} = (I_3 + J_3)/8, \quad \det H = 128.$$

For $x \in X$, put $y_i = \langle x, t_i \rangle$ and write $f(y)$ for the number of vectors with that signature. The projection norm is $y^t H^{-1} y \leq 6$. The certificate also supplies integer rows W with $WGT^t = I_3$. Hence the section is primitive: if an ambient lattice vector has the form $x = aT$, then $a = xGW^t$ is integral. The section has eight norm-six vectors: the three basis vectors, their sum, and their negatives. Completeness follows because a section vector of norm six has integer coefficients bounded in absolute value by $\sqrt{6(H^{-1})_{ii}} = \sqrt{3/2} < 2$.

Unless x is one of those eight vectors, the ambient minimum gives $|y \cdot z| \leq 3$ for each of their section coefficient rows z . The eight exceptional signatures have multiplicity one. Enumeration over the integer box $[-6, 6]^3$, with the norm and root inequalities, therefore gives a complete domain of 239 signatures. Removing the eight exceptions and identifying antipodes leaves 116 nonnegative unknown fibre multiplicities, including the zero signature.

For a multi-index α of degree $2m \leq 10$, the design identity is

$$\sum_{x \in X} \prod_{i=1}^3 \langle x, t_i \rangle^{\alpha_i} = \frac{|X|6^m}{48 \cdot 50 \cdots (48 + 2m - 2)} \sum_P \prod_{(a,b) \in P} H_{i_a i_b}, \quad (10.1)$$

where P runs over pairings of the multiset with α_i copies of i . Subtracting the exceptional signatures gives $Av = b$ with 161 moment rows and 116 variables. An antipodal nonzero orbit

has coefficient $2y^\alpha$; the zero orbit has coefficient 0^α . All entries are rational and are regenerated from H .

Let c be the coefficient vector of

$$f(e_1) + f(e_2) + f(e_3) - 9f(-2, -2, 3).$$

The rational dual certificate provides a multiplier λ , with 107 nonzero entries, satisfying

$$c - A^t \lambda \geq 0, \quad \lambda^t b = 7377408. \quad (10.2)$$

Because $v \geq 0$, this proves

$$f(e_1) + f(e_2) + f(e_3) \geq 7377408 + 9f(-2, -2, 3). \quad (10.3)$$

Thus the moments bound the desired statistic even when they leave individual fibre sizes undetermined. The checker reconstructs the whole domain and all moment entries, then directly checks the stored multiplier and every rational reduced cost in (10.2).

10.3. The witness and the kissing configuration. The certificate supplies 298 distinct integer coefficient rows w with

$$wGw^t = 6, \quad TGw^t = (-2, -2, 3)^t.$$

Consequently $f(-2, -2, 3) \geq 298$. Project the shell vectors in the three fibres as follows.

Theorem 10.2. *The projections of the shell vectors with signatures e_1, e_2, e_3 , normalized to unit length, form a kissing configuration in dimension 45 with at least 7380090 points.*

Proof. Project the three fibres e_1, e_2, e_3 to the section's orthogonal complement. Each projected squared norm is $6 - 1/4 = 23/4$. Two original distinct vectors have inner product at most three. Their projected products are therefore at most $3 - 1/4 = 11/4$ within one fibre and $3 - 1/8 = 23/8$ in different fibres. This also excludes collisions. Normalization gives a kissing configuration in dimension 45, and (10.3) yields

$$K(45) \geq 7377408 + 9(298) = 7380090.$$

The 298-vector witness and the moment inequality together establish the count. $\square$

10.4. The change of search representation. For a squared-norm-eight anchor p , define $A_p = \{x \in X : \langle x, p \rangle = 4\}$. Integrality and the minimum bound imply $|\langle x, p \rangle| \leq 4$. Applying the eighth moment to $P(t) = t^2(t^2 - 1)(t^2 - 4)(t^2 - 9)$ gives

$$2P(4)|A_p| = 90961920, \quad P(4) = 20160, \quad |A_p| = 2256.$$

This converts completeness of a proposed parent set into a finite check of 2256 distinct valid vectors. Relations between endpoints can then be evaluated as common-neighbour counts on the same parent set, replacing repeated lattice enumeration for each section.

The earlier Pless-lattice route found 292 witnesses. The 298-vector construction instead uses an anchor in the odd-coordinate sector of the QR neighbour: in numerator coordinates it has shape $(3^6, 1^{42})$ up to signs and permutation. Pair each $x \in A_p$ with $p - x$. The resulting 1128 pairs are centred at $p/2$. Join two pairs when representatives have inner product one or three; changing an endpoint replaces the product by four minus itself, so adjacency is well defined. The graph has degree 368, and a selected nonadjacent pair represented by r, s has 70 common neighbours.

For each of the $368 - 70 = 298$ neighbours of r that are not neighbours of s , orient its endpoint a so that $\langle r, a \rangle = 3$. Then $\langle s, a \rangle = 2$. With section rows $T = (-s, s - p, p - r)$, the vector $w = p - a$ has squared norm six and signature $(-2, -2, 3)$. The parent certificate specifies the anchor, chosen endpoints and complete parent set; its inputs are listed in Table 8. The checker reconstructs this graph and verifies that the extracted section and 298-vector set equal the final certificate. The additional search freedom lies in the parent anchor's sector and shape.

11. FINITE SEARCH AND MATHEMATICAL STRUCTURE

Qiushi Engine's investigation connected changes in geometric representation with finite optimization and counting problems. For replacements, the objective is the net gain after all required deletions. For lattice sections, the relevant object is a linear statistic of fibre multiplicities. These reductions retain the constraints that determine the final point count.

11.1. An exact exchange objective. Let $X \subset S^{d-1}$ be a kissing configuration and let $C \subset S^{d-1} \setminus X$ be a finite candidate set. For $c \in C$, write $F(c) = \{x \in X : \langle c, x \rangle > 1/2\}$. Join two candidates when they are incompatible. For an independent set $I \subset C$,

$$X_I = (X \setminus F(I)) \cup I, \quad F(I) = \bigcup_{c \in I} F(c), \quad |X_I| - |X| = |I| - |F(I)|. \quad (11.1)$$

The retained old points are compatible, every old–new conflict has been removed, and the selected new points are pairwise compatible. Thus maximizing this expression over a candidate pool is an exact finite subproblem of the geometric search.

For binary variables z_c and r_x indicating insertion and removal, respectively, the corresponding integer program is

$$\max \left(\sum_{c \in C} z_c - \sum_{x \in X} r_x \right), \quad z_c + z_{c'} \leq 1 \ (cc' \in E), \quad r_x \geq z_c \ (x \in F(c)).$$

At an optimum, $r_x = 1$ precisely on the union of the selected conflict sets. When candidates and deletions are disjoint bundles of internally compatible points, their cardinalities become the coefficients in the objective. This is the weighted form behind a support exchange, a signed replacement, and a first/second-layer head selection.

Write $g(I) = |I| - |F(I)|$. If $c \notin I$ is compatible with I , then

$$g(I \cup \{c\}) - g(I) = 1 - |F(c) \setminus F(I)|.$$

Thus a candidate pays only for blockers not already removed. Its marginal gain can increase as the selected family grows, which is why screening candidates by their individual insertion costs can discard profitable exchanges. The final length-33 witness inserts ten supports with nine distinct blockers, despite twenty blocker incidences.

11.2. When the candidate representation changes. A finite candidate pool fixes the coordinates of every candidate. Its conflict graph therefore describes those coordinates. A coordinate change can alter an edge, remove a blocker, or admit a previously excluded point. The geometric lifting criterion (2.3) gives the conditions to rebuild the graph after such a change.

The motion in Section 3, the signed replacements in Section 6, and the added shell in dimension 38 change this finite problem rather than merely improve its incumbent solution. The two optimizations in dimensions 49–55 also interact: image families chosen for one line class need not remain optimal after the class grows. Equation (8.4) separates the resulting gains.

The moment calculations use a different compression. They retain all admissible projection signatures and optimize or identify a linear statistic of their multiplicities. Exact dual multipliers then express that statistic in terms of known moments. For dimension 43 the target is determined by an equality; for dimension 45 a nonnegative slack gives a lower bound. The proof requires completeness of the signature domain, but not prior knowledge of each individual fibre.

11.3. Further structural questions. The constructions leave several precise questions. Which tail Gram matrices permit the equality case in the reuse bound while supporting larger head classes? Which boundary conditions allow a signed local code to attain Proposition 6.4? For embedded sections, which parent root systems force a child count independently of the ambient lattice, as in Theorem 9.1? The last question is distinct from optimizing the determinant of the

section: the two isometric D_5 sections in Section 9 already have different perpendicular shell populations.

12. FINITE VERIFICATION

Each lower bound has two parts: a mathematical reduction covering all types of point pairs, and a finite witness satisfying the reduction’s hypotheses. The supplement contains these witnesses and programs for checking the required identities and inequalities. Table 8 identifies the finite objects used in the proofs; paths are relative to `constructions/`.

TABLE 8. Finite inputs to the construction proofs.

Dimensions	Mathematical input	Location
25	Head data, the 552-point layer and two repair vectors	<code>d25/</code>
27	Five 496-point blocks; first-layer heads and 311 labelled second-layer owners	<code>leech/</code> , <code>d27/</code>
32–34, 37, 39	Complete weight-eight support lists, parent families and exchanges	<code>codes/data/</code>
35–37	Signed local codes and their boundary embeddings	<code>codes/</code>
38	Shell partition, tail rotation and 144 added lines	<code>d38/</code>
49–55	The 7077-line class, isometries and tail assignments	<code>p48/</code>
43	Parent embedding, orbit data and target-count certificate	<code>sections/d43/</code>
45	QR-neighbour basis, moment multipliers, parent graph and 298 witnesses	<code>sections/qr/</code> , <code>sections/d45/</code>

For dimension 25, the checker first certifies the 197579-point input from its integer and rational head data. The 552-point motion has symbolically preserved equalities, as proved in Section 3. The remaining strict inequalities use 192-bit outward-rounded Arb intervals or integer comparisons after squaring positive quantities. The two repair vectors are given as integer rows with exact normalization.

For the triangular Leech lifting, the inputs are five 496-point blocks, the Golay coordinate convention, and exact tail coordinates. Integer products, membership, disjointness and radical identities establish the hypotheses of the lifting theorem. The separate implementations are `leech/verify.py` and `leech/verify_independent.py`. The updated dimension-27 construction additionally checks all first-layer owners, the 311 second-layer owners and labels, and every interlayer inequality in Section 4.4. For the layered constructions, support intersection checks are combined with the distances of the dense binary code or its coset unions. The five supplied parent families also reproduce the support exchanges, including the exact union of deleted conflicts. The signed replacements additionally check every new signed pair and every boundary support intersection; in dimension 37 the checker regenerates all 512 vectors from the 32 transverse four-subsets.

For the bounds in dimensions 49–55, the checker verifies the 7077-line mother class in the catalogue Gram basis, the exact image matrices, the seven union counts, and an injective assignment to antipodal tail lines. The published minimum and the theta-series identity supply the mother-shell premise. The same images applied to the original 7069-line class verify every term in Table 7. The ambient minimum supplies the cross-block inequality.

For dimension 38, the checker regenerates the Leech shell from the base package’s Golay generator, checks the class partition, tail triples and rational rotation, and verifies the 144 new norm-48 lines against both the complete shell and one another. All pair products in the additional tests are integers. Two implementations check the 144-line extension.

The dimension-43 verifier checks the concrete lattice embedding, then regenerates the 26641-signature domain, subgroup action, moment matrix and target vector from standard E_8 coordinates. It checks the supplied rational multiplier against this regenerated system. These latter steps use only the hypotheses of Theorem 9.1. The dimension-45 checks reconstruct the specified QR neighbour, verify its exact basis Gram, establish the norm-four parity obstruction, and check all 298 fibre witnesses. The parent-graph calculation checks all 2256 endpoints, their pairing about $p/2$, and the extraction of the same 298 witnesses. A separate rational calculation regenerates the 161-row moment system and checks the supplied multiplier in (10.2) directly.

12.1. Autonomous mathematical research. The nineteen constructions arose from Qiushi Engine's autonomous mathematical research, encompassing construction searches, mathematical analysis and computational verification. Its research connected changes of representation with new finite objects: coordinated layer motion (Section 3), joint label selection (Section 4.4), signed-design transfer (Section 6), and moment functionals adapted to the section counts (Sections 9 and 10). The accompanying research accounts describe these mathematical developments and identify their inputs and resulting constructions. Background on the research system is given in [13].

APPENDIX A. THE INTEGER LEECH SHELL AND TAIL CONVENTIONS

This appendix fixes the normalization of the classical Golay description of the Leech shell [2]. Let $G \subset \mathbb{F}_2^{24}$ be generated by the twelve words in Table 9, with coordinate zero at the left. Enumeration and binary elimination give

$$\dim G = 12, \quad G \subseteq G^\perp, \quad W_G(z) = 1 + 759z^8 + 2576z^{12} + 759z^{16} + z^{24}. \quad (\text{A.1})$$

An *octad* is the support of a weight-eight word of G . Two distinct octads intersect in at most four places, because their symmetric difference is a nonzero word of weight at least eight. For $c \in G$ put $\varepsilon_i(c) = (-1)^{c_i}$ and define

$$\Sigma_A = \{4\varepsilon e_i + 4\varepsilon' e_j : 0 \leq i < j < 24, \varepsilon, \varepsilon' \in \{\pm 1\}\}, \quad (\text{A.2})$$

$$\Sigma_B = \left\{ 2 \sum_{i \in O} \eta_i e_i : O \text{ an octad}, \eta_i \in \{\pm 1\}, \prod_{i \in O} \eta_i = 1 \right\}, \quad (\text{A.3})$$

$$\Sigma_C = \left\{ \sum_{i=0}^{23} \varepsilon_i(c) e_i - 4\varepsilon_j(c) e_j : c \in G, 0 \leq j < 24 \right\}. \quad (\text{A.4})$$

Set $\Sigma = \Sigma_A \cup \Sigma_B \cup \Sigma_C$.

Lemma A.1. *This set satisfies (4.1).*

Proof. The absolute coordinate patterns distinguish the three families. Support and signs recover the parameters of the first two. In the third, the unique coordinate of absolute value three identifies j , after which the signs recover c . Thus the cardinalities are

$$|\Sigma_A| = 4 \binom{24}{2} = 1104, \quad |\Sigma_B| = 759 \cdot 128 = 97152, \quad |\Sigma_C| = 4096 \cdot 24 = 98304.$$

They sum to 196560. The respective squared norms are $16 + 16$, $8 \cdot 4$, and $9 + 23$, all equal to 32.

For two A -vectors, different supports give at most one contribution of 16, and equal supports with different signs give at most zero. For A - B pairs there are at most two contributions of eight; for A - C pairs the bound is $4(3 + 1) = 16$. For B - B pairs on different octads, the intersection bound gives at most $4 \cdot 4 = 16$. On the same octad, different even sign patterns differ at least twice, giving at most $4(8 - 4) = 16$.

For a B - C pair, let O be the octad and j the exceptional coordinate of the C -vector. If $j \notin O$, the inner product is at most $2|O| = 16$. Otherwise put $y_i = \eta_i \varepsilon_i(c)$ on O . Self-orthogonality of G and the even sign condition imply $\prod_{i \in O} y_i = 1$, and the inner product is

$$2 \sum_{i \in O} y_i - 8y_j.$$

If $y_j = 1$, this is at most eight. If $y_j = -1$, at least one other y_i is negative, so the sum is at most four and the expression is at most sixteen.

Finally consider C - C pairs with words c, c' and exceptional coordinates j, k . Set $s_i = (-1)^{c_i + c'_i}$ and $S = \sum_i s_i = 24 - 2 \text{wt}(c + c')$. If $j = k$, the inner product is $S + 8s_j$; distinct vectors then require $c \neq c'$, giving $S \leq 8$. If $j \neq k$, it is $S - 4(s_j + s_k)$. This equals sixteen when $c = c'$, and is at most $8 + 8$ otherwise. These six family pairs exhaust all cases. $\square$

TABLE 9. A binary generator for the Golay code in (A.1). Coordinates run from zero to 23 from left to right.

Row	Binary word
0	010111100101000000000001
1	110111001010000000000010
2	100011011101000000000100
3	000110111110000000001000
4	011101011100000000010000
5	111001101001000000100000
6	001001110111000001000000
7	011010001111000010000000
8	101100011011000100000000
9	101111110000001000000000
10	110010110011010000000000
11	111100100110100000000000

For compatibility with the supplied tail arrays, label V by

$$\begin{array}{c|ccc|ccc}
 i & & & & i & & & \\
 \hline
 0 & -1 & 0 & -1 & 6 & -1 & 0 & 1 \\
 1 & 0 & -1 & -1 & 7 & 1 & -1 & 0 \\
 2 & -1 & -1 & 0 & 8 & 0 & -1 & 1 \\
 3 & 0 & 1 & -1 & 9 & 1 & 1 & 0 \\
 4 & -1 & 1 & 0 & 10 & 0 & 1 & 1 \\
 5 & 1 & 0 & -1 & 11 & 1 & 0 & 1.
 \end{array} \tag{A.5}$$

The triangles in (4.6) then have labels $\{0, 7, 10\}$, $\{1, 6, 9\}$, $\{2, 3, 11\}$, and $\{4, 5, 8\}$. For the pure tails set

$$(\sigma(0), \dots, \sigma(11)) = (0, 3, 4, 1, 5, 2, 9, 6, 10, 7, 8, 11), \quad p_a = Rv_{\sigma(a)}/\sqrt{2}.$$

The source rows $\hat{t}_i, \hat{p}_a$ in `constructions/leech/data/tails.json` are related to these coordinates by right multiplication by

$$O = \begin{pmatrix} \sqrt{2}/2 & 0 & \sqrt{2}/2 \\ -\sqrt{6}/6 & \sqrt{6}/3 & \sqrt{6}/6 \\ -\sqrt{3}/3 & -\sqrt{3}/3 & \sqrt{3}/3 \end{pmatrix}, \quad O^T O = I. \tag{A.6}$$

In row-vector notation, $\hat{t}_i O = v_i^\top / \sqrt{6}$ and $\hat{p}_a O = v_{\sigma(a)}^\top R^\top / \sqrt{2}$. The exact coordinate-change test checks these identities for every label, so the common rotation preserves both within-set and cross-set inner products.

APPENDIX B. BINARY GENERATORS AND COSET REPRESENTATIVES

Table 10 gives the generators used for the three dense sign codes. An entry is a hexadecimal mask, representing the binary word c by $\sum_i c_i 2^i$. The generator of length 32 is used unchanged in ambient dimensions 32–37. The length-38 and length-39 columns are the kernels reconstructed from (5.9). Table 11 supplies the affine representatives; the length-32 code uses only zero.

TABLE 10. Generator masks for H_{32} , H_{38} , and H_{39} . Blank entries reflect their different dimensions.

Row	H_{32}	H_{38}	H_{39}
0	4EF0C081	2000C0DE0B	4000152727
1	24A09082	1000AA229D	2000C0DE0B
2	48609084	8009F5CD6	1000AA229D
3	1E505088	4004FAE6B	8009F5CD6
4	6C905090	200ED9AAD	4004FAE6B
5	FA30C0A0	100BC80CE	200ED9AAD
6	AAC000C0	805E4067	100BC80CE
7	D2608100	40E56DAB	805E4067
8	7E60D200	20B8FB4D	40E56DAB
9	B8601400	1096303E	20B8FB4D
10	D800D800	84B181F	1096303E
11	CC606000	4EFC197	84B181F
12	1EE10000	2BDAD53	4EFC197
13	B2B20000	1949B31	2BDAD53
14	74740000		1949B31
15	D8D80000		
16	FF000000		

TABLE 11. Representatives of the eleven cosets in length 38 and the ten cosets in length 39.

a	r_a in length 38	r_a in length 39
0	0	0
1	C6E94	ADCFB
2	DF6ABF	F7795F
3	27B441	740E29
4	D3CE66	940DFF
5	41BBBB	3B66AB
6	BC1E1F	6CD301
7	52727	4709D4
8	9951C8	98A926
9	7001EB	D23688
10	3F6969	

The complete kernel weight distributions are given in Table 12. The least nonzero weights give the within-coset distance bounds. For every distinct pair of representatives in Table 11, the affine

space $r_a + r_b + H_m$ has minimum weight ten. Enumeration of these spaces establishes both the cross-coset distances and disjointness of the cosets. This determines each code in Proposition 5.3 entirely from the printed data.

TABLE 12. Weight distributions $A_w = |\{h \in H_m : \text{wt}(h) = w\}|$. Every omitted weight has coefficient zero.

w	H_{32}	H_{38}	H_{39}
0	1	1	1
8	908	0	0
10	3328	0	0
12	14784	669	970
14	27392	0	0
16	38246	5289	8952
18	27392	0	0
20	14784	8002	16452
22	3328	0	0
24	908	2310	5988
28	0	113	402
32	1	0	3

APPENDIX C. AN EXPLICIT PLESS NEIGHBOUR AND ITS MINIMAL SHELL

This appendix establishes the minimum and shell size in an explicit integer model of an even neighbour. The theta-series argument applies to every extremal even unimodular lattice of rank 48, including the catalogue lattice in Section 8 and the QR neighbour in Section 10.

C.1. The specified even neighbour. Index the rows and columns of a 24×24 matrix S by $\{\infty\} \cup \mathbb{F}_{23}$, in that order. With χ the quadratic character modulo 23, set

$$S_{\infty\infty} = 0, \quad S_{\infty j} = 1, \quad S_{i\infty} = -1, \quad S_{ij} = \chi(i - j) \quad (i, j \in \mathbb{F}_{23}). \quad (\text{C.1})$$

The identity $SS^T = 23I$ follows from the quadratic-character sum $\sum_x \chi(i - x)\chi(j - x) = -1$ for $i \neq j$. Consequently the ternary code

$$C_3 = \{(u, uS) : u \in \mathbb{F}_3^{24}\} \subset \mathbb{F}_3^{48}$$

is self-dual: its systematic generator has rank 24, and $I + SS^T = 24I = 0$ over $\mathbb{F}_3$. Put

$$M = \{z \in \mathbb{Z}^{48} : z \bmod 3 \in C_3\}, \quad L = M/\sqrt{3}. \quad (\text{C.2})$$

The row basis

$$\begin{pmatrix} I_{24} & S \\ 0 & 3I_{24} \end{pmatrix}$$

has determinant 3^{24} . The code orthogonality makes L integral, and the scaling gives covolume one. It is odd, since it contains $3e_0/\sqrt{3}$ of squared norm three.

Let $L_0 = \{x \in L : x \cdot x \in 2\mathbb{Z}\}$ and define

$$c = (1^{24}, -23, 1^{23}), \quad g = c + 6e_0, \quad L' = L_0 \cup \left(\frac{g}{2\sqrt{3}} + L_0 \right). \quad (\text{C.3})$$

The characteristic congruence $c \cdot z/3 \equiv z \cdot z/3 \pmod{2}$ for $z \in M$ is checked on the displayed basis. This suffices since the squared norm modulo two is additive on an integral lattice. Also $g/\sqrt{3} \in L_0$, $g \cdot g/3 = 208$, and $g/(2\sqrt{3})$ has squared norm 52. Its product with $z/\sqrt{3} \in L_0$ is

integral by the characteristic congruence. Thus (C.3) is an even lattice, obtained by extending the index-two sublattice L_0 by two cosets, and has covolume one.

The matrix (C.1) is the defining matrix of the Pless symmetry code $C(23)$, with the same signs and coordinate order as [36, Section 1]. In particular, C_3 has minimum Hamming weight 15 [36, 37]. The sum of the rows of $(I \mid S)$ is c , so C_3 contains the all-one word. We shall also use the following consequence of the complete weight enumerator of an extremal ternary self-dual [48, 24, 15] code containing that word:

$$\sum_{b \in C_3 \cap \{0,1\}^{48}} t^{\text{wt}(b)} = 1 + 94t^{24} + t^{48}. \quad (\text{C.4})$$

This is the two-symbol specialization of [35, equation (20)]. These two code-theoretic facts determine which of the possible short vectors can occur in the even neighbour.

Theorem C.1 (The mother shell). *The lattice L' in (C.3) is even unimodular, has minimum squared norm 6, and has exactly 52416000 vectors of squared norm 6.*

Proof. Evenness and unimodularity were established above. Suppose first that $z/\sqrt{3} \in L_0$ has squared norm 2 or 4. Then $z \cdot z$ is 6 or 12. A nonzero residue $z \bmod 3$ would have Hamming weight at least 15, whereas its weight is at most $z \cdot z$. Hence $z = 3w$ for some integer vector w . Its squared norm is $3w \cdot w$, which is even only when $w \cdot w$ is even, and is therefore at least 6 if nonzero.

A vector in the other coset has the form $y/(2\sqrt{3})$, where $y = g + 2z$ and $z/\sqrt{3} \in L_0$. All 48 coordinates of y are odd, so its squared norm is at least 4. Equality would force $y \in \{1, -1\}^{48}$. Let b be the incidence vector of its negative coordinates and put $w = \text{wt}(b)$. Since $g \equiv \mathbf{1} \pmod{3}$, the identity $z = (y - g)/2$ gives $-z \equiv b \pmod{3}$. Consequently $b \in C_3$, and (C.4) implies $w \in \{0, 24, 48\}$. On the other hand, $c_i \equiv 1 \pmod{6}$, $\sum_i c_i = 24$, and $c \cdot g = 582$. It follows that

$$c \cdot z = \frac{\sum_i c_i - c \cdot g}{2} - \sum_{i: b_i=1} c_i \equiv 3 - w \equiv 3 \pmod{6}. \quad (\text{C.5})$$

This contradicts $z/\sqrt{3} \in L_0$, which requires $c \cdot z \equiv 0 \pmod{6}$. Thus neither coset contains a vector of squared norm 2 or 4. The vector $\sqrt{3}(e_0 + e_1) \in L_0$ has squared norm 6, proving the asserted minimum.

It remains to count the minimum shell. Write $\Theta_{L'}(q) = \sum_{x \in L'} q^{x \cdot x/2}$. The theta series of an even unimodular lattice of rank 48 is a modular form of weight 24 [2, 38]. The space has basis $E_4^6, E_4^3 \Delta, \Delta^2$. Its constant term and the vanishing coefficients of q and q^2 therefore give

$$\Theta_{L'} = E_4^6 - 1440E_4^3 \Delta + 125280\Delta^2. \quad (\text{C.6})$$

Substituting $E_4 = 1 + 240q + 2160q^2 + 6720q^3 + O(q^4)$ and $\Delta = q - 24q^2 + 252q^3 + O(q^4)$ yields

$$\Theta_{L'} = 1 + 52416000q^3 + O(q^4),$$

as required. $\square$

Within this Pless-code model, the shift can equivalently be written using $g_* = \mathbf{1} + 6e_0$: indeed, $(g - g_*)/2 = -12e_{24} \in \sqrt{3}L_0$, so g and g_* define the same coset of L_0 . This is the all-one neighbour construction in [35]; the parity calculation above explains directly why this choice excludes the norm-four vectors. The liftings of Section 8 use the catalogue Gram basis and its published minimum directly; the projection of Section 10 uses the separately specified QR neighbour.

The norm-six shell of L' , divided by $\sqrt{6}$, is a kissing configuration C of size 52416000. Indeed, for two distinct shell vectors x, x' , the nonzero lattice vector $x - x'$ has squared norm at least six, so $x \cdot x' \leq 3$.

APPENDIX D. FLEXIBILITY OF AUXILIARY CONFIGURATIONS

The layer inequalities identify which constraints are tight and which leave room for deformation. A *contact* is a pair of distinct points with inner product $1/2$. Here X_{25} is the 197058-point endpoint model of (4.2), and X_T, X_M, X_R are the ERS layers of Section 5. The dimension-38 observation concerns that layered-code construction; the 591900-point construction instead uses the Leech lift of Section 7. The connection between contact constraints and rigidity of spherical codes is developed in [7]; here it can be made quantitative directly from the proved inner-product margins.

D.1. Paired motion in dimension 25. Write $s \in S_*/\sqrt{32}$ and $x_s^\pm = (\sqrt{3}s/2, \pm 1/2)$, with poles $p^\pm = (0, \pm 1)$. The contacts involving these new points are exactly the 496 paths

$$p^+ \text{ --- } x_s^+ \text{ --- } x_s^- \text{ --- } p^-. \quad (\text{D.1})$$

Their internal vertices are disjoint. They have no contacts with the retained equator, because the relevant inner products are at most $\sqrt{3}/4 < 1/2$. Between different directions the same-latitude bound is $7/16$, and the opposite-latitude bound is $-1/16$. Thus the displayed graph has 994 vertices and 1488 edges.

Proposition D.1. *Keep the equator and the two poles of X_{25} fixed. For each $s \in S_*/\sqrt{32}$ choose independently a unit vector $\hat{s}$ with $\|\hat{s} - s\| < 1/24$. Replacing $x_s^\pm$ by $(\sqrt{3}\hat{s}/2, \pm 1/2)$ preserves the kissing condition and the contacts (D.1).*

Proof. Let $\delta = \max_s \|\hat{s} - s\| < 1/24$. For unit vectors,

$$|\langle \hat{s}, \hat{t} \rangle - \langle s, t \rangle| \leq \|\hat{s} - s\| + \|\hat{t} - t\| \leq 2\delta.$$

Consequently distinct same-latitude directions have inner product less than $7/16 + 3\delta/2 < 1/2$, and distinct opposite-latitude directions have inner product less than $-1/16 + 3\delta/2 < 0$. For a fixed equatorial direction z ,

$$\frac{\sqrt{3}}{2} \langle z, \hat{s} \rangle \leq \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{2} \delta < \frac{13\sqrt{3}}{48} < \frac{1}{2}.$$

The last inequality follows from $3 \cdot 13^2 < 24^2$. A paired pair still has inner product $1/2$, as do its contacts with the appropriate poles. All other pole products are unchanged. $\square$

With the equator, poles, and labels fixed, the common direction of each pair is free to move in an open subset of S^{23} . The height is forced, but the direction is not. Thus saturation of the latitude constraints does not imply rigidity of the lifted configuration.

D.2. Independent motion of the dense layers. In dimensions $m = n = 38, 39$, the dense sign codes have minimum distance ten. Their mutual inner products are therefore at most $1 - 20/m < 1/2$. Their products with the middle and root layers are at most $\sqrt{8/m}$ and $\sqrt{2/m}$, also strictly less than $1/2$. Every dense point is an isolated vertex of the contact graph.

Proposition D.2. *For $m = n \in \{38, 39\}$, keep X_M and X_R fixed and put*

$$\eta_m = \frac{40 - m}{4m}.$$

Each $x \in X_T$ may independently be replaced by a unit vector $\hat{x}$ with $\|\hat{x} - x\| < \eta_m$. The resulting configuration has the same cardinality and satisfies the kissing condition. No new contact involves a dense point.

Proof. The change in a product of two unit vectors is at most the sum of their displacement lengths. Hence distinct moved dense points have inner product strictly less than

$$1 - \frac{20}{m} + 2\eta_m = \frac{1}{2}.$$

For a moved dense point and a fixed middle point, the bound is $\sqrt{8/m} + \eta_m < 1/2$. To verify the latter inequality exactly, the positive number $1/2 - \eta_m$ has square exceeding $8/m$ by $153/5776$ at $m = 38$ and by $937/24336$ at $m = 39$. The root-layer bound is smaller. The other pairs are unchanged. $\square$

The respective displacement radii are $1/76$ and $1/156$. The codes provide exact starting configurations, while their neighbourhoods need not retain a binary description. The two motion results exhibit distinct sources of flexibility: collective motion of a contacting pair in dimension 25, and independent motion of strictly separated points in dimensions 38 and 39.

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